# Who Pays to Compete? Wealth, a Fixed Fee and Entry by Rank

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## 1 Introduction

### 1.1 Research question

A credential is often worth more as a token than as evidence of skill. An institution that allocates scarce positions, and cannot cheaply observe the quality it cares about, has reason to weight a standardised token that it can verify at little cost: a credential is a cleaner status than the disorganised status that comes from unverified providers. Once the token can be bought at a fixed price, who holds it depends on who can afford it. Two current institutions show the pieces. Chiba City allocates licensed daycare places, within priority tiers, to the highest selection points, and adds four points for a family that uses an unlicensed nursery or a babysitter for at least 64 hours a month; families without such care still compete on their other points. Italian contracting authorities must award extra points in public tenders to firms that hold a gender-equality certificate, which attests a firm's employment policies against parameters set by decree; firms without it still bid and can win. In the first the purchase is a service that earns standing, and in the second the authority weights a certificate that attests what it does not check itself. The sources are in `lit/applications_2026`.

The question is what this interaction of credentialism and wealth produces on its own. A fixed fee buys a component of standing in a contest for positions awarded by rank; outsiders still compete on the base component, and wealth matters only through the utility cost of the fee. How do the number who enter, and the entrants' share of the positions, respond to the prize, the fee, a spread of wealth, the number of challengers, and the weight the rule puts on the base component? Accumulated skill and saving, the channels through which wealth usually works, are absent by construction, so the answers isolate what the purchasable token does. Why an institution chooses the weight is outside the model, which takes it as given; the account above motivates the weight and is not derived here. Weberian closure, in which outsiders cannot win, is the boundary at which the base component counts for nothing. The model keeps apart the two things Weber [1978] separates as class and status. Wealth sets only the cost of entering, and the prize is won by rank (rows L14 to L16 give the passages and where the parallel ends).

### 1.2 Main answer

The answer rests on the headline claims below, one lever at a time. Each is a statement about the one-shot equilibrium at a given wealth profile and holder of the position (Appendix D, "The contest").

- **The count and who enters.** Every pure-strategy equilibrium has the same number of entrants, and the richest of them entering is always an equilibrium (K1). Sorting by wealth needs only that the fee's utility cost falls with wealth, which strict concavity implies and which is weaker (K3).
- **The prize and the fee.** A larger prize does not lower the count, a larger fee does not raise it (K5), and the prize caps entry for every number of challengers (K4).
- **A spread of wealth.** A spread about a pivot moves the count in a direction set by one comparison, the marginal entrant against the pivot (K2). This sign rule is new among the papers read (C1, C5).
- **The number of challengers.** As challengers are added, the first entrant's gain can rise and then fall, and never falls and then rises (K6).
- **The weight on the base component.** A rule that weighs the base component more draws no more entrants (K8) and gives them no larger share of the prize (K7), a sign that is new among the papers read (C9, C12). It

does not thereby pick a better winner. With the entrants fixed the winner's base component does not fall, but once entry responds a common bonus, such as Chiba City's four points, can leave the winner with a lower base component (K9).

- **The entrants' share.** The share of the prize taken by the entrants is set by the count and does not fall as more enter; when entry buys nothing it is $\frac{k+1}{Q+1}$ with $k$ entrants (K7). As a function of the count it is new among the papers read (C13).

## 1.3 Headline claims

Table 1: Headline claims, each traced to the model rows it rests on.

| ID | Headline | Model rows | Scope note |
|---|---|---|---|
| K1 | Every pure-strategy equilibrium, with an indifferent challenger entering, has the same number of entrants, $k^*$, and the $k^*$ richest challengers entering is always one of them. | M5, M6, M7, M8, M26, M27, M29, M30 | The count is pinned. Who enters is pinned when the condition of M7 holds, and need not be otherwise. The count rests on a gain that depends on the number of entrants alone (M26); for some laws of the bought component that differ by identity, falling with wealth (M27) or rising with it (M29), and some costs, it is not pinned. An incumbent who competes under the same rule, taken to be the lowest-cost player, sits out while a challenger enters only when $g(k^*) < \kappa_0$ and $\kappa_{k^*} \leq g(k^*-1)$ (M30). At exact ties, with an indifferent challenger staying out, other counts can occur. |
| K2 | The effect of a pivot-spread of wealth on how many enter has a sign set by one comparison. Entry weakly rises when the marginal entrant stands at or above the pivot, and weakly falls when it stands at or below. | M12, M13, M14, M15 | Proved for pivot-spreads (M13). For any change of profile, a two-rank comparison signs the change outside the band (M12, M14); inside the band the signs of the change settle nothing (M14), and the mean cannot replace the pivot (M13). A strict fall is shown only for linear spreads with a large enough stretch, when the fee's utility cost diverges at the fee (M15). |
| K3 | Given score laws that do not depend on who enters, the sorting needs only that the fee's utility cost falls with wealth, which strict concavity implies and which is strictly weaker. | M5, M8, M23, M24, M26 | No condition on $u'''$ is used. When the law of the bought component differs by identity the sorting can fail (M27). Wealth sorting into a tournament is in print, by example (Lazear and Rosen), under declining absolute risk aversion, which needs $u''' > 0$, so the mechanism is not claimed as new. Hu, Immorlica and Vaughan derive a group cost condition the same way, the same outlay weighing more in the disadvantaged group's utility (L33). |
| K4 | Entry is capped by the prize for every number of challengers. An index $m$ with $\kappa_m \leq \Delta(m)$ has $(m+1)\kappa_m \leq V$, and every member $i$ of an equilibrium $S$ has $\lvert S\rvert\, \kappa_i \leq V$. | M10 | The accounting behind the cap is in print for identical agents (Fu, Jiao and Lu; Fu and Lu). It replaces the old mobility ceiling and is not claimed as new. |
| K5 | A larger prize does not lower the number of entrants, and a larger fee does not raise it. | M11 | Weak monotonicity only. |
| K6 | When the score gap is single-peaked, the first entrant's gain can rise and then fall in the number of challengers, and it never falls and then rises. With a uniform base component it never falls, for some base components it must eventually fall, and stochastic dominance alone does not sign it. | M16, M17, M18, M19, M20, M21, M22 | The absence of an interior minimum is Ryvkin and Drugov's Karlin step applied to the gain, with the score gap in place of the noise density, and the message that no such prediction is universal is theirs (L8, C12). What is new among the papers read is the uniform case and, for a bought component of two values, a condition on the base component that makes the gap single-peaked (C7). |

| ID | Headline | Model rows | Scope note |
|---|---|---|---|
| K7 | The entrants take the prize with a probability that does not fall as more of them enter. With $k$ entrants it equals $\frac{k+1}{Q+1}$ when entry buys nothing, and, with costs fixed, it does not rise as the rule weighs the base component more. | M28, M31 | The share counts positions and not their pay; earnings after entry are outside the model (Appendix D). Its direction in the weight on the base component is that of the count (K8). The entrants' share as a function of how many enter is new among the papers read (C13). |
| K8 | With the fee, the prize and the wealth profile fixed, the number of entrants does not rise as the rule weighs the base component more, and neither does the gain from entering at any number of entrants. | M28 | Proved with the incumbent scored as an entrant, for every atomless law of the base component, every law of a non-negative bought component and every number of challengers. For an incumbent protected from the rule, whose law does not move with the weight, the argument does not apply (Appendix D). The sign is weak, and its size is not derived. In the two noise-comparative entry models read the sign is the opposite, because effort rises with the accuracy of the ranking and competes away the rents of entry, while a fixed fee does not rise (C9). The sign is new among the papers read (C9, C12). |
| K9 | With the entrants fixed, a rule that weighs the base component more never leaves the winner with a lower base component. Once entry responds it can, as with a common bonus. | M32 | The first part holds pointwise, for any laws and any bought scores, for weights up to one. The second is shown by one example, with one challenger, a uniform base component and a common bonus; how large the fall can be and how often it occurs are not derived. A common bonus washes out when every player holds it and favours its holders once some stop buying (K8). The result is a scope on K8, not a claim of novelty. |

## 2 The model in brief

**Summary.** One prize of fixed value $V$ goes to the highest score among an incumbent and $Q$ challengers. Challengers differ in wealth, and every challenger knows the whole wealth profile. Each challenger makes one choice, whether to enter by paying a fee $c$ that is the same for all. A challenger's score is $\sigma_i = \mu r_i + (1-\mu)s_i e_i$. The base component $r_i$ is drawn for everyone, and the bought component $s_i \geq 0$ counts only for an entrant ($e_i = 1$). The rule's weight on the base component is $\mu$. An outsider still competes, on the base component alone. Both components are drawn after the choices, from laws that do not depend on wealth, so no challenger knows her own score when choosing. The incumbent is scored as an entrant. A challenger enters when the gain from entering, $\Delta(m)$ with $m$ other entrants, covers the utility cost of the fee, $\kappa(w) = u(w) - u(w-c)$. The cost falls with wealth, and the gain falls as more others enter. These two facts pin the number of entrants, $k^*$, in every pure-strategy equilibrium, and the $k^*$ richest entering is always an equilibrium.

Table 2: The players, their choices and what each optimises.

| Player | Who it is | Choice | Objective | Known when choosing |
|---|---|---|---|---|
| Challenger $i$ | One of $Q$, $i \in \{0,\ldots,Q-1\}$, with wealth $w_i$, indexed from the richest. One with $w_i \leq c$ does not enter. | Enter or not, $e_i \in \{0,1\}$. No effort and no amount to bid. | The one-period expected payoff $u(w_i - c\,e_i) + V\Pr(i \text{ wins})$, given the others' choices; a repeated contest is this game if players are myopic or value holding the position alike (Appendix D). | $V$, $c$, $\mu$, the laws of $r$ and $s$, the whole wealth profile. Not her own draws. |

| Player | Who it is | Choice | Objective | Known when choosing |
|---|---|---|---|---|
| Incumbent | The holder of the position, scored as an entrant ($C = F$). | None in M1 to M29, M31 and M32, where she always competes and holds the bought component; whether she paid for it enters no result, since her payoff does not. In M30, enter or not, paying $c$ like a challenger, as player 0 with the lowest cost. | In M30, the same as a challenger's. | As a challenger. |
| Nature | | Draws $r_i$ and $s_i$, independent of each other, across players and of wealth. | None. | |
| The institution | Not a player. | Fixes $V$, $c$ and $\mu$, which the comparative statics move one at a time. | None in the model; why it chooses $\mu$ is outside the model. | |

**Entry, equilibrium and timing.** Entering changes a challenger's expected payoff by $\Delta(m) - \kappa(w_i)$. Here $m$ is the number of other entrants, and $\Delta(m)$ is $V$ times the rise in the probability of winning that entry buys (M1). A challenger therefore enters exactly when $\kappa(w_i) \leq \Delta(m)$. The equilibrium is a pure-strategy Nash equilibrium in the entry choices, an entrant set $S$ with $\kappa_i \leq \Delta(|S|-1)$ for every entrant and $\Delta(|S|) < \kappa_j$ for every outsider (M6). The wealth profile is realised first and known to all challengers. Challengers then choose simultaneously, and entrants pay $c$. Nature draws the two components, and the highest score takes the prize. The game is played once.

**Comparative statics.** How the equilibria respond to the prize, the fee, a spread of wealth, the number of challengers and the weight on the base component is the answer of Section 1.2, and the headline claims of Section 1.3 give each response with its scope.

# 3 Model

Table 3: Results of the model, each with its statement and the Lean theorems that check it.

| ID | Claim | Statement | Lean | Proof |
|---|---|---|---|---|
| M1 | Representation (P1). | $\Delta(m) = V\int \varphi\, dH_m = V\,\mathbb{E}[\varphi(M_m)]$, with $M_m \sim H_m$. | `EntryContestP1P2.two_max_sum`, `EntryContestP1P2.p1_representation`, `EntryContestP1P2.Delta_eq_expectation` | p. 24 |
| M2 | Dominance (P2). | For $s \geq 0$ and $\mu \leq 1$, $F \leq G$ and $\Delta(m) \geq 0$. For $\mu < 1$ and $\Pr(s > 0) > 0$, $F < G$ on some $[q,u)$ with $q < u$. | `EntryContestP1P2.p2_fosd`, `EntryContestP1P2.p2_strict_interval`, `EntryContestP1P2.Delta_nonneg_from_primitives` | p. 26 |
| M3 | Step identity (P3). | For $m+2 \leq Q$, $\Delta(m+1) - \Delta(m) = -\frac{V}{2}\int \varphi^2\, dK_m$. | `EntryContestStep.three_max_sum_real`, `EntryContestStep.step_identity`, `EntryContestModel.Delta_step` | p. 26 |

| ID | Claim | Statement | Lean | Proof |
|---|---|---|---|---|
| M4 | Monotonicity. | $\Delta(m+1) \le \Delta(m)$ for $V \ge 0$, with $<$ when $V > 0$ and $\int \varphi^2 \, dK_m > 0$. | `EntryContestModel.Delta_step_nonpos`, `EntryContestModel.DeltaUpTo_step`, `EntryContestEq.Delta_step_strict` | p. 26 |
| M5 | Threshold entry (P5). | For $\kappa_0 \le \cdots \le \kappa_{Q-1}$ and $\Delta$ non-increasing, $\kappa_j \le \Delta(j)$ and $i \le j$ give $\kappa_i \le \Delta(i)$. | `EntryContest.single_crossing`, `EntryContestModel.p5_threshold_from_primitives` | p. 27 |
| M6 | Count invariance. | Every equilibrium $S$, with $\kappa_i \le \Delta(\lvert S\rvert - 1)$ for $i \in S$ and $\Delta(\lvert S\rvert) < \kappa_j$ for $j \notin S$, has $\lvert S\rvert = k^*$. | `EntryContestEq.count_invariance_fin`, `EntryContestMM.exists_mem_ge`, `EntryContestMM.exists_not_mem_le` | p. 27 |
| M7 | Identities pinned. | If $\Delta(k^* - 1) < \kappa_{k^*}$, the assortative set $\{0, \ldots, k^* - 1\}$ is the only equilibrium. | `EntryContest.members_below_kstar`, `EntryContestEq.assortative_unique`, `EntryContestEq.assortative_only`, `EntryContestEq.assortative_only_model` | p. 27 |
| M8 | Assortative selection (N4). | $\{0, \ldots, k^* - 1\}$ is an equilibrium; $\sum_{i<k} \kappa_i \le \sum_{i \in S} \kappa_i$ for $\lvert S\rvert = k$; with identical score laws, payoff sums of equal-size equilibria differ by their cost sums. | `EntryContestEq.assortative_is_equilibrium`, `EntryContestEq.assortative_min_cost`, `EntryContestEq.prize_term_anonymous`, `EntryContestEq.payoff_gap_anonymous` | p. 27 |
| M9 | Benchmark (P6). | With the incumbent scored as an entrant and $s \ge 0$ atomless, $\Delta(m) = V/(m+2)$ at $\mu = 0$ for every $Q$, and $\Delta(m) \to V/(m+2)$ as $\mu \to 0$. | `EntryContestP6P7.integral_iid`, `EntryContestP6P7.integral_cdf_pow`, `EntryContestP6P7.p6_evaluation`, `EntryContestP6P7.p6_at_zero`, `EntryContestP6P7.tendsto_cdf_investorLaw`, `EntryContestP6P7.tendsto_cdf_nonInvestorLaw`, `EntryContestP6P7.p6_limit` | p. 28 |
| M10 | Saturation (P7). | $\Delta(m) \le V/(m+1)$ for every $Q$; an index $m$ with $\kappa_m \le \Delta(m)$ has $(m+1)\kappa_m \le V$, and every member $i$ of an equilibrium $S$ has $\lvert S\rvert \, \kappa_i \le V$. | `EntryContestP6P7.integral_cdf_pow`, `EntryContestP6P7.p7_Delta_le`, `EntryContestP6P7.p7_cap`, `EntryContestP6P7.p7_count` | p. 28 |

| ID | Claim | Statement | Lean | Proof |
|---|---|---|---|---|
| M11 | Prize and fee (P8). | With challengers at or below the fee out, $k^*$ is non-decreasing in $V$ for $\mu \leq 1$ and $s \geq 0$, and non-increasing in $c$ for $u$ non-decreasing on $(0, \infty)$. | `EntryContestSpread.count_ge_of_enters`, `EntryContestSpread.count_mono_gain`, `EntryContestSpread.count_mono_prize`, `EntryContestEq.Delta_mono_prize`, `EntryContestMM.Delta_scale`, `EntryContestSpread.count_antitone_fee` | p. 28 |
| M12 | Margin condition. | For sorted profiles $w$ and $w'$, $w_{k^*-1} \leq w'_{k^*-1}$ gives $k^* \leq k'^*$, and $w'_{k^*} \leq w_{k^*}$ gives $k'^* \leq k^*$. | `EntryContestSpread.enters_down`, `EntryContestSpread.count_ge_of_enters`, `EntryContestSpread.count_le_of_fails`, `EntryContestSpread.margin_rise`, `EntryContestSpread.margin_fall` | p. 29 |
| M13 | Pivot spreads (P9-gen, P9). | For a pivot-spread $T$ about $x_0$, $k^*$ weakly rises when $w_{k^*-1} \geq x_0$ and weakly falls when $w_{k^*-1} \leq x_0$. The linear spread about the mean is one, and keeps the mean. A change that keeps the total and majorizes $w$ can lower $k^*$ with $w_{k^*-1}$ above the mean, for every small $\mu > 0$. | `EntryContestSpread.pivot_rise`, `EntryContestSpread.pivot_fall`, `EntryContestSpread.spread_isPivot`, `EntryContestMM.mm2a`, `EntryContestSpread.mps_lowers_count`, `EntryContestSpread.mps_lowers_witness_counts`, `EntryContestSpread.mps_lowers_count_positive` | p. 29 |
| M14 | The band. | With $d = w' - w$, the signs of $d$ leave the direction of $k^*$ open only when $d_{k^*-1} < 0 < d_{k^*}$. A pivot-spread never does so. Inside the band two changes that keep the total and majorize $w$, with the same signs at every rank, move $k^*$ in opposite directions for every small $\mu > 0$. | `EntryContestSpread.branch_of_not_band`, `EntryContestSpread.not_band_of_single_crossing`, `EntryContestSpread.pivot_single_crossing`, `EntryContestSpread.mps_band_signs`, `EntryContestSpread.mps_witness_counts`, `EntryContestSpread.mps_band_both_directions`, `EntryContestSpread.mps_band_both_directions_positive` | p. 29 |

| ID | Claim | Statement | Lean | Proof |
|---|---|---|---|---|
| M15 | Strictness under a spread (P9). | If some entrant lies below $x_0$ at $\lambda = 1$ and $\kappa(w) \to \infty$ as $w \downarrow c$, $k^*(\lambda)$ is non-increasing in $\lambda \geq 1$ and $k^*(\lambda) < k^*(1)$ above some $\bar{\lambda}$. | `EntryContestKappa.exit_below_pivot`, `EntryContestKappa.enters_downward`, `EntryContestKappa.p9_strict`, `EntryContestKappa.p9_monotone`, `EntryContestKappa.p9_strict_model` | p. 30 |
| M16 | Competitor number (P-MU). | For $Q \geq 1$ and every $C$, $D(Q) = -V\left(\int W_Q\, dF - \int W_Q\, dG\right)$, so for $V > 0$, $D(Q) > 0$ exactly when $\int W_Q\, dF < \int W_Q\, dG$. The kernel $g^n(1-g)$ rises on $[0, \frac{n}{n+1}]$ and falls on $[\frac{n}{n+1}, 1]$. | `EntryContestPMU.weight_eq`, `EntryContestPMU.pmu_identity`, `EntryContestPMU.pmu_sign_iff`, `EntryContestPMU.kernel_rises`, `EntryContestPMU.kernel_falls`, `EntryContestPMU.kernel_max` | p. 30 |
| M17 | Rise then fall in $Q$. | If $dF - dG \leq 0$ on $(-\infty, x_0]$ and $dF - dG \geq 0$ on $(x_0, \infty)$, and $V \geq 0$, then $D(Q_2) \leq G(x_0)^{Q_2 - Q_1} D(Q_1)$ for $1 \leq Q_1 \leq Q_2$, and $\min\left(\Delta_{Q_1}(0), \Delta_{Q_3}(0)\right) \leq \Delta_{Q_2}(0)$ for $1 \leq Q_1 \leq Q_2 \leq Q_3$. | `EntryContestPMU.weight_shift`, `EntryContestPMU.pmu_single_crossing`, `EntryContestPMU.pmu_step_nonpos_persists`, `EntryContestPMU.quasiconcave_of_steps`, `EntryContestPMU.pmu_quasiconcave` | p. 31 |
| M18 | Uniform base component. | For $r$ uniform on $[0,1]$, $s \geq 0$, $0 < \mu \leq 1$, $V \geq 0$ and every $C$, the hypothesis of M17 holds at $x_0 = \mu$, and $D(Q) \geq 0$ for $Q \geq 1$, so $\Delta_Q(0)$ is non-decreasing in $Q \geq 1$. | `EntryContestPMU.uniform_left`, `EntryContestPMU.uniform_right`, `EntryContestPMU.pmu_quasiconcave_uniform`, `EntryContestRefute.pmu_step_nonneg_uniform`, `EntryContestRefute.Delta_mono_uniform` | p. 31 |
| M19 | A class with rise then fall. | For $s \in \{0,1\}$ with $\Pr(s=0) = p$, $0 < \mu < 1$, $V \geq 0$, $\Pr(r \leq 0) = 0$ and $\Pr(r \in B) \leq \Pr(r \in B - c)$ whenever $c > 0$ and $B \subseteq (c, \infty)$, the hypothesis of M17 holds at $x_0 = 1 - \mu$, and $\Delta_Q(0)$ is quasi-concave in $Q \geq 1$ for every $C$. A density of $r$ non-increasing on $(0, \infty)$ gives the shift condition, and the smaller of two uniform draws has one. | `EntryContestShift.left_of_lift`, `EntryContestShift.right_of_lift`, `EntryContestShift.quasiconcave_of_shiftMono`, `EntryContestShift.shiftMono_of_density`, `EntryContestShift.lawA_shiftMono`, `EntryContestShift.witness_quasiconcave` | p. 31 |

| ID | Claim | Statement | Lean | Proof |
|---|---|---|---|---|
| M20 | The fall, exhibited. | For $r$ the smaller of two uniform draws, $\mu = \frac{3}{4}$, $s$ as in M19 and $C = F$, $D(1)$ and $D(2)$ are quadratics in $p$ (in the proof). For $V > 0$, $p = \frac{1}{2}$ gives $\Delta_1(0) < \Delta_2(0)$ and $\Delta_3(0) < \Delta_2(0)$, and $p = \frac{1}{4}$ gives $\Delta_2(0) < \Delta_1(0)$. | `EntryContestFall.stepQ_one`, `EntryContestFall.stepQ_two`, `EntryContestFall.rise_then_fall`, `EntryContestFall.fall_from_one` | p. 32 |
| M21 | When the gain must fall. | With $s$ and $r$ as in M19, $t = 1-\mu$ and $G_t$ the law of $\mu r + t$, if $t < a < b$, $G(t) < G(a)$, $G(b) < 1$, $C(a) > 0$ and $G\big((a,b]\big) < G_t\big((a,b]\big)$, then for $V > 0$ and $p < 1$ some $Q_0$ has $D(Q) < 0$ for every $Q \geq Q_0$. The base law of M20 meets this on $(\frac{1}{2}, \frac{5}{8}]$ for every $p < 1$, and a uniform base component never does. | `EntryContestMustFall.must_fall_general`, `EntryContestMustFall.shift_dominates`, `EntryContestMustFall.stepQ_bern`, `EntryContestMustFall.must_fall`, `EntryContestMustFall.witness_must_fall`, `EntryContestMustFall.unif_shiftMono`, `EntryContestMustFall.uniform_never_gains` | p. 32 |
| M22 | Dominance does not sign $D(Q)$. | With $C = F$, the pairs $(F,G) = \big(x,\, 1-(1-x)^2\big)$ and $(x^2, x)$ on $[0,1]$ both have $F \leq G$ and no atoms. The first has $D(1) = \frac{V}{60}$ and $D(2) = \frac{V}{420}$, and the second has $D(Q) = -\frac{VQ}{(Q+2)(Q+3)(Q+4)}$. | `EntryContestPMUWitness.pairA_fosd`, `EntryContestPMUWitness.pairB_fosd`, `EntryContestPMUWitness.pairA_step_one`, `EntryContestPMUWitness.pairA_step_two`, `EntryContestPMUWitness.pairB_step`, `EntryContestPMUWitness.fosd_does_not_sign` | p. 32 |
| M23 | Burden-monotonicity. | $\kappa(w) = u(w) - u(w-c)$ is strictly decreasing on $w > c$ when $u$ is strictly concave on $(0,\infty)$, or when $u'(w) < u'(w-c)$ for every $w > c$. A convex stretch of $u$ longer than $c$ rules it out. Log and CRRA utility ($\gamma > 0$, $\gamma \neq 1$) satisfy it. | `EntryContestKappa.burden_strictAnti_of_strictConcave`, `EntryContestBM.burden_strictAnti_of_deriv`, `EntryContestBM.not_burden_of_convex_stretch`, `EntryContestKappa.log_burden_strictAnti`, `EntryContestKappa.crra_strictConcave`, `EntryContestKappa.crra_burden_strictAnti` | p. 33 |
| M24 | Weaker than concavity. | For every $c > 0$ some strictly increasing $u$, not concave on $(0,\infty)$, has $\kappa$ strictly decreasing on $(c,\infty)$ and $\kappa(w) \to \infty$ as $w \downarrow c$. The parameters of the construction form an open set, and its convex stretch can have any length below $c$. | `EntryContestBM.rampU_burden`, `EntryContestBM.rampU_kappa_diverges`, `EntryContestBM.rampU_strictConvex`, `EntryContestBM.rampU_not_concave`, `EntryContestBM.admissibleParams_open`, `EntryContestBM.rampU_admissible`, `EntryContestBM.admissible_width`, `EntryContestBM.bm_strictly_weaker` | p. 33 |

| ID | Claim | Statement | Lean | Proof |
|---|---|---|---|---|
| M25 | Divergence at the fee. | For $u$ continuous at $c$, $\kappa(w) \to \infty$ as $w \downarrow c$ exactly when $u(x) \to -\infty$ as $x \downarrow 0$. Log utility and CRRA with $\gamma > 1$ satisfy it, and CRRA with $\gamma < 1$ does not. | `EntryContestKappa.kappa_diverges_iff`, `EntryContestKappa.not_diverges_of_continuousWithinAt_zero`, `EntryContestKappa.log_kappa_diverges`, `EntryContestKappa.crra_kappa_diverges`, `EntryContestKappa.crra_kappa_not_diverges` | p. 33 |
| M26 | Anonymity. | If every entrant's score has the law $F$, challenger $i$'s gain from entering against the entrants $T$ is $\Delta(\lvert T\rvert)$, whoever they are. | `EntryContestAnon.profile_mem`, `EntryContestAnon.profile_not_mem`, `EntryContestAnon.cdf_profile`, `EntryContestAnon.gain_anonymous` | p. 34 |
| M27 | The anonymity boundary. | If the law of $s_j$ varies with $j$, the gain can depend on who the other entrants are. With the bought component falling in wealth, an equilibrium of one entrant and an equilibrium of two can coexist for every small $\mu > 0$: with $V = 1$, $r$ uniform on $[0,1]$, $s_j$ the largest of 1, 2 and 3 uniform draws, the incumbent's of one, and $\kappa$ from $u(x) = -1/x$, $c = 1$, $w = (\frac{11}{4}, \frac{9}{4}, 2)$. | `EntryContestAnon.winProb_zero_entrant`, `EntryContestAnon.winProb_zero_outsider`, `EntryContestAnon.gain_zero`, `EntryContestAnon.kappa3_values`, `EntryContestAnon.gains3`, `EntryContestAnon.anon_witness_zero`, `EntryContestAnon.gain_tendsto`, `EntryContestAnon.anon_witness_positive`, `EntryContestAnon.anon_witness_exists` | p. 34 |
| M28 | Weight on the base component. | With the incumbent scored as an entrant, $s \geq 0$, $r$ atomless and $V \geq 0$, for $0 < \mu' \leq \mu$ and every $Q$ and $m$, $\Delta(m;\mu) \leq \Delta(m;\mu')$, and with fixed costs $k^*$ is non-increasing in $\mu$. | `EntryContestMu.Delta_normalise`, `EntryContestMu.cdf_payerLaw_antitone`, `EntryContestMu.first_integral`, `EntryContestMu.Delta_payer_mono`, `EntryContestMu.Delta_antitone_mu`, `EntryContestMu.kstar_antitone_mu` | p. 34 |

| ID | Claim | Statement | Lean | Proof |
|---|---|---|---|---|
| M29 | The boundary, aligned. | With the bought component rising in wealth, an equilibrium of one entrant and an equilibrium of two can coexist at $\mu = 0$ and for every small $\mu > 0$: with $V = 1$, $r$ uniform on $[0,1]$, $s_j$ the largest of 12, 10 and 9 uniform draws, the incumbent's of one, and $\kappa$ from $u(x) = -1/x$, $c = 1$, $w = (\frac{43}{20}, \frac{52}{25}, \frac{83}{40})$. | `EntryContestAligned.kappaA_values`, `EntryContestAligned.ability_with_wealth`, `EntryContestAligned.costs_ordered`, `EntryContestAligned.gainsA`, `EntryContestAligned.aligned_witness_zero`, `EntryContestAligned.aligned_witness_positive`, `EntryContestAligned.aligned_witness_exists`, `EntryContestAligned.control_window` | p. 35 |
| M30 | The incumbent as a challenger. | Let the incumbent be player 0 of a game of $Q+1$ players with $\kappa_0 \le \kappa_j$, and let $g(m)$ be the gain against $Q$ rivals of whom $m$ are in: $\Delta(m)$ at $C = G$ for $m \le Q-1$, and $\Delta(Q-1)$ at $C = F$ for $m = Q$. Then $g$ is non-increasing, her being in gives a challenger $g(m+1)$ for every $m \le Q-1$, and every equilibrium has $k^*$ entrants, with $k^*$ the count of this game. An equilibrium with 0 out and some challenger in exists iff $1 \le k^* \le Q$, $g(k^*) < \kappa_0$ and $\kappa_{k^*} \le g(k^*-1)$; $\{0,\dots,k^*-1\}$ is an equilibrium in every case, and if $g(k^*-1) < \kappa_{k^*}$ then 0 is in every non-empty equilibrium. | `EntryContestInc.Delta_incumbent_in_eq`, `EntryContestInc.symGain_step`, `EntryContestInc.Delta_incumbent_in_incGain`, `EntryContestInc.incGain_top_step`, `EntryContestInc.incGain_antitone`, `EntryContestInc.incumbent_out_iff`, `EntryContestInc.incumbent_out_iff_model`,`EntryContestInc.incumbent_out_not_unique`, `EntryContestInc.incumbent_assortative_model`,`EntryContestInc.incumbent_in_of_pinned`, `EntryContestInc.incumbent_in_of_pinned_model`, `EntryContestInc.incumbent_count_model`, `EntryContestInc.band_witness`, `EntryContestInc.control_no_band` | p. 35 |

| ID | Claim | Statement | Lean | Proof |
|---|---|---|---|---|
| M31 | The entrants' share. | With the incumbent scored as an entrant and $k$ of $Q$ challengers entering, the prize goes to an entrant with probability $\xi(k) = \int G^{Q-k}\, d(F^{k+1})$. For $F$ atomless, $\xi$ is non-decreasing in $k$ and $\xi(Q) = 1$; for $F = G$, $\xi(k) = \frac{k+1}{Q+1}$; for $s \geq 0$ and $r$ atomless, $\xi(k)$ is non-increasing in $\mu > 0$, and with fixed costs so is $\xi(k^*)$. | `EntryContestShare.payerShare_full`, `EntryContestShare.payerShare_eq_one_sub`, `EntryContestShare.payerShare_monotone`, `EntryContestShare.payerShare_at_equal`, `EntryContestShare.payerShare_normalise`, `EntryContestShare.payerShare_antitone_mu`, `EntryContestShare.payerShare_kstar_antitone_mu`, `EntryContestShare.control_share_needs_nonneg` | p. 36 |
| M32 | Selection in the weight. | With the entrants fixed, if $i$ has the highest score at $\mu_1$ and $j$ at $\mu_2$, $\mu_1 < \mu_2 \leq 1$, then $r_i \leq r_j$, so the winner's expected base component does not fall in $\mu$. With entry responding it can fall: for $Q = 1$, $r$ uniform on $[0,1]$, $s \equiv 1$, $V = 1$ and the incumbent scored as an entrant, $\Delta(0) = \frac{1}{2}$ at $\mu = \frac{1}{2}$ and $\frac{3}{8}$ at $\mu = \frac{2}{3}$, a challenger with $\kappa = \frac{7}{16}$ enters at $\frac{1}{2}$ and not at $\frac{2}{3}$, and the winner's expected base component is strictly lower at $\frac{2}{3}$. | `WeightSelection.winner_base_mono`, `WeightSelection.winner_base_mono_expect`, `WeightSelection.gain_half`, `WeightSelection.gain_two_thirds`, `WeightSelection.enters_at_half`, `WeightSelection.out_at_two_thirds`, `WeightSelection.selection_falls_with_exit`, `WeightSelection.winner_base_mono_needs_le_one` | p. 36 |
| M33 | Implications for the rule. | A fee that differs across challengers can reverse the order of their costs. At a fee with $(k+1)\kappa_k > V$, at most $k$ challengers enter. A compression of wealth toward $x_0$ weakly raises $k^*$ when $w_{k^*-1} \leq x_0$ and weakly lowers it when $w_{k^*} \geq x_0$. With $s \geq 0$, $r$ atomless and $0 < \mu \leq 1$, $\xi(k) \geq \frac{k+1}{Q+1}$, and $\xi(k) \neq \frac{k+1}{Q+1}$ implies $F \neq G$. | `Implications.varying_fee_reorders`, `Implications.cap_sets_count`, `Implications.cap_sets_count_model`, `Implications.compression_rises`, `Implications.compression_falls`, `Implications.compression_rises_needs_below`, `Implications.share_off_benchmark`, `Implications.share_ge_benchmark` | p. 37 |

# 4 Literature

No paper read signs how a pivot-spread of wealth moves the entry count. Here, given anonymity and an entry cost that falls with wealth, the marginal entrant against the pivot sets the sign, and outside the band two ranks suffice where Costrell and Loury use one quantile fixed by technology. Every equilibrium has the same entry count, new for finitely many challengers whose entry costs differ with wealth, because the gain depends on the count alone, and the assortative set settles who enters without the devices of models with identical entrants. The first entrant's gain has the form of Ryvkin and Drugov's marginal benefit, and that it never falls in the number of challengers

with a uniform base component is new. With the incumbent scored as an entrant, the entry count does not rise in the weight on the base component, a new sign, opposite to the effort models read. The entrants' share of the prize as a function of the entry count is new, since elsewhere outsiders cannot win or agents form a continuum, and it does not rise in the weight on the base component. Only an entrants' share above its benchmark, the entry count plus one over the number of challengers plus one, shows that the bought component counts, which bears on Tilly's opportunity hoarding.

Table 4: Papers the model or the conclusions depend on, each with a verbatim quote.

| ID | Paper | What we rely on | Quote | Page | Model rows |
|---|---|---|---|---|---|
| L1 | Lazear and Rosen [1981] | Wealth sorting into a tournament is in print, by example and through declining absolute risk aversion, a condition on $u'''$. M23 sorts through $\kappa$ alone, which $u'' < 0$ gives. | "Individuals will self-select the payment scheme in accordance with their wealth." | 855 | M5, M23 |
| L2 | Levin and Smith [1994] | With identical potential entrants, pure strategies leave who enters unexplained, and mixed entry makes the count random. | "The process by which potential bidders divide into these two groups is not explained" | 586 | M6, M7 |
| L3 | Morgan et al. [2012] | With identical players the entrants' identities are not pinned, and sequential entry orders them. | "the identity of the players choosing to opt into the contest is not uniquely determined" | 441 | M6, M7 |
| L4 | Moreno and Wooders [2011] | Under private entry costs every buyer compares her cost with one threshold, and the count is binomial. Here the threshold depends on the index. | "indicating the maximum entry cost for which the buyer enters the auction" | 319–320 | M5, M6 |
| L5 | Fullerton and McAfee [1999] | With heterogeneous types the efficient entry equilibrium admits the lowest-cost firms. Profits depend on rivals' types, and entry equilibria of sizes 2 and 3 coexist (`FullertonMcAfee.two_sizes`), so count invariance needs anonymity. | "entry of the m lowest-cost firms is an equilibrium" | 580 | M6, M7, M26 |
| L6 | Fu et al. [2015] | The rent-accounting bound behind M10, for identical agents with mixed entry, capping an expected count. | "the rent to be dissipated in the contest cannot exceed" | 397 | M10 |
| L7 | Fu and Lu [2010] | Total effort equals the budget less the entrants' total entry cost, in a contest chosen by a designer. *Unverified:* the working paper (MPRA 945) was read, and the *Economic Inquiry* version is not checked. | "the equilibrium total effort is given by the difference between the total budget of the contest organizer and the total entry costs incurred by participating contestants, regardless of the contest technology." | 10 | M10 |

| ID | Paper | What we rely on | Quote | Page | Model rows |
|---|---|---|---|---|---|
| L8 | Ryvkin and Drugov [2020] | The order-statistic kernel, and that no prediction in the number of players holds for every noise law. M16 to M22 are the entry-gain instance. | "There is not a single prediction, either for individual or aggregate effort, that cannot be reversed for at least some distribution of noise." | 1589 | M16, M17, M22 |
| L9 | Costrell and Loury [2004] | A spread's effect signed by the position of one marginal quantile, which is fixed by technology there and endogenous in M12. | "completely characterize the technology" | 1327 | M12, M13 |
| L10 | Hopkins and Kornienko [2010] | The dispersive order as the order for inequality comparisons, quoting Shaked and Shanthikumar (2007). | "A variable with distribution $F$ is said to be smaller in the dispersive order (or less dispersed) than a variable with a distribution $G$" | 128 | M13, M14 |
| L11 | Lewis and Thompson [1981] | The same order, defined earlier by quantile spacing, which with a crossing point gives the pivot-spreads of M13. | "We call the relationship defined by (1.1), (1.2) an ordering in dispersion of $G$ and $H$" | 77 | M13, M14 |
| L12 | Schroyen and Treich [2016] | The privilege-contest payoff, with a prize valued apart from wealth, and an intensive-margin effect of a spread that turns on $u'''$. *Unverified:* the working paper (No. 16-699) was read, and the *Games and Economic Behavior* version is not checked. | "and therefore its marginal value is independent of the level of wealth" | 8 | M13, M23 |
| L13 | Cole et al. [1992] | A more compact income distribution raises savings through matching on wealth. The direction matches the below-pivot branch of M13, and the mechanism differs. | "the agent from the economy with the more compact income distribution will tend to have a higher savings rate" | 1103 | M13 |
| L14 | Weber [1978] | Property excludes the poor from competing, and does so through marginal utility. Here that channel is the fee's utility cost $\kappa$, which falls with wealth (M23), so entry is richest-first (M5). | "The mode of distribution, in accord with the law of marginal utility, excludes the non-wealthy from competing for highly valued goods" | 927 | M5, M23 |
| L15 | Weber [1978] | Class and status are distinct, and money does not by itself confer status. Here wealth sets only the cost of entry, and the prize goes by rank on draws that do not depend on wealth. Weber gives this anonymity to the market and not to the status order (p. 936), so the contest is market-like. | "Money and an entrepreneurial position are not in themselves status qualifications, although they may lead to them" | 306 | M26, M27 |

| ID | Paper | What we rely on | Quote | Page | Model rows |
|---|---|---|---|---|---|
| L16 | Weber [1978] | Closure, participation subject to a condition. The fee is such a condition, and M4 gives entrants an interest in it, since each further entrant lowers the gain. No agent sets the fee here. | "participation of certain persons is excluded, limited, or subjected to conditions" | 43 | M4, M10 |
| L17 | Laurison and Friedman [2016] | Access to the traditional professions is more closed than to management, which earns more, so closure does not follow the prize. Their main finding, a pay gap after entry, is outside the model, which stops at entry. *Unverified:* the accepted version was read, and the published pagination is not checked. | "the higher professions remain significantly more elitist in terms of restricting access for those from working class backgrounds" | 13 | M5, M11, M28 |
| L18 | Wegener [1992] | A survey of prestige in stratification theory, which separates closure from a hierarchy of positions and finds no theory that joins them. The question here is narrower, namely how wealth differences decide who competes for positions open to every entrant on one score. | "None of these theories is able to solve the problem of how theoretically to merge the idea of social closure with that of a hierarchy of positions." | 253 | M5, M26 |
| L19 | Tilly [1998] | In opportunity hoarding a bounded set keeps access to a valuable resource. The outcome is the same here, the richest $k^*$ taking the prize with probability $\xi(k^*)$. The means differ, since in Tilly the network acts and here no one does. *Unverified:* one page read, and the edition is not confirmed. | "If members of a network acquire access to a resource that is valuable, renewable, subject to monopoly, supportive of network activities, and enhanced by the network's modus operandi, network members regularly hoard access to the resource, creating beliefs and practices that sustain their control." | 154 | M8, M31 |
| L20 | Morgan et al. [2022] | With an effort choice, contestants drop out as ranking of the costly choice becomes more accurate, once effort competes the rents away. Their accuracy is a fall in $\mu$ here, so the sign is the opposite of M28; entry by a fixed fee has no effort margin. *Unverified:* the 2018 working paper was read, and the *Journal of Economic Theory* version is not checked. | "at a critical point, competition becomes so intense that contestants start 'dropping out,' i.e., they put in zero effort" | 4 | M28 |

| ID | Paper | What we rely on | Quote | Page | Model rows |
|---|---|---|---|---|---|
| L21 | Shaked [1982] | The dispersive order holds exactly when every shift of one distribution function crosses the other at most once, from below (Theorem 2.1). This is the crossing in the pivot-spreads of M13. He credits the relation to Saunders and Moran (1978) and its name to Lewis and Thompson (L11). | "In the first pair of theorems it is shown that order in dispersion is equivalent to the condition that some functions, determined by the pair of the underlying distributions, change sign at most once." | 311 | M13, M14 |
| L22 | Drugov and Ryvkin [2020] | Effort falls as noise becomes more dispersed, for any prize schedule (Proposition 1), the sign of M28. With free entry, the less noisy tournament has weakly fewer entrants (Section 5.3), the opposite sign, because effort rises as noise falls. *Unverified:* the working paper (NES WP 256) was read, and the *Journal of Economic Theory* version is not checked. | "the dispersive order is a necessary and sufficient condition for the ranking of equilibrium efforts in general" | 10 | M28 |
| L23 | Mathews and Namoro [2008] | Two events pay different prizes, and a strong and a weak contestant each join one. The strong contestant can join the event with the smaller prize and the weak one the event with the larger prize (their allocation c), so a smaller prize can attract the stronger field. In M11 a larger $V$ never lowers the number who pay the fee, because $V$ raises the gain from entering at every count. Their contestants choose between events at no entry cost, the number of strategic contestants is fixed at two, and wealth does not appear. *Unverified:* the working paper of 29 March 2005 was read, and the published version (*Journal of Economics* 2008) is not checked. | "Event 2 is able to attract a field of higher quality by offering smaller prizes." | 13 | M8, M11 |

| ID | Paper | What we rely on | Quote | Page | Model rows |
|---|---|---|---|---|---|
| L24 | Corcoran [1984] | Free entry into a Tullock lottery among identical players: players enter while the return on the lottery exceeds the return on other projects and exit while it falls short. At the long-run count the total spent equals the discounted prize, and for $r > 1$ the count $r/(r-1)$ does not depend on the prize. Expenditure in his model is proportional to the prize. Here the fee $\kappa$ does not depend on $V$, $k^*$ is non-decreasing in $V$ (M11), and an entrant at index $m$ has $(m+1)\kappa \leq V$ (M10). His players are identical, their number is a real number, and wealth does not appear. | "If the rate of return obtainable on other projects is less than that offered by the rent-seeking option then players will enter; if greater, players will exit." | 91 | M5, M10, M11 |
| L25 | Fryer and Loury [2013] | Agents choose a binary investment in skills whose cost is drawn from their group's distribution; productivity is a noisy function of the investment, and a fixed measure of slots is sold to the highest bidders, so those who do not invest still compete. The laissez-faire investment rate is unique, the investors are the agents below a cost threshold, and the group with higher costs invests and holds slots at a lower rate. This is the class gap of M31 for groups, in a continuum with priced slots; there is no count of payers, no fixed prize and no weight on talent. Fryer and Loury [2010], the working paper, conjectures that a blind rule lowers investment in both groups, and the published version drops the conjecture. *Unverified:* the journal's page proofs of 8 June 2013 were read, and the final pagination is not checked. | "With no policy intervention the disadvantaged will be less than proportionally represented among slot owners, compared to their presence in the population, since agents face a common slots price in market equilibrium." | 3 | M5, M6, M31, C6, C10, C12, C13 |

| ID | Paper | What we rely on | Quote | Page | Model rows |
|---|---|---|---|---|---|
| L26 | Fu and Lu [2009] | Identical contestants choose a continuous investment that lowers their cost of effort, and then all compete in a Tullock contest. The paper finds investment and effort non-monotonic in the number of contestants. Within the symmetric pure-strategy equilibria it studies, that does not hold: at its parameter zero investment is not an equilibrium for $K \geq 4$, and its candidate for $K = 3$ never is one (record in `lit/fu_lu_2009`). There is no binary investment and no one who does not invest, so nothing corresponds to the gain of M17 and M18. *Unverified:* the authors' manuscript of February 2009 was read, and the *Economics Letters* pages are not checked. | "Positive equilibrium investment emerges when the number of contestants is small, while zero investment occurs in equilibrium when the number of contestants is large." | 2 | M17, M18, C7 |
| L27 | Fryer and Loury [2003] | Workers with private costs exert effort in pair-wise contests for a fixed wage, and under laissez-faire the group with higher costs wins fewer contests. Under a group-blind redistribution rule the unique D1 equilibrium pools the highest-cost workers at zero effort; they still compete, winning against each other at random, and the pool grows as the redistribution target becomes more aggressive. This is the direction of M28, for a redistribution target, with a continuous effort and no fee. *Unverified:* the working paper (NBER WP 10104) was read, and the *Journal of Economic Inequality* version (2005) is not checked. | "And, this pool increases with the aggressiveness of the redistributive goal." | 18 | M28, C12 |

| ID | Paper | What we rely on | Quote | Page | Model rows |
|---|---|---|---|---|---|
| L28 | Coate and Loury [1993] | Workers choose a binary investment whose cost is private and has the same law in both groups. A noisy test reads the investment, and employers assign to the better job by a threshold they set from their beliefs, so a worker who did not invest can still get it. The investment rate need not be unique, because the standard responds to the belief. In their uniform example, at a fixed employer policy, their formulas imply that both investment rates fall as the test becomes less accurate; the paper does not state this, and the derivation is ours (record in `lit/coate_loury_1993`). Under an equal-assignment rule employers can patronize a group by lowering its standard, which lowers its investment. With unequal cost laws the group with higher costs does less well. There is no fixed prize, no rank and no fee, and the number assigned moves with the standard. | "Lowering the standard may reduce investment incentives, however, because the favored workers see themselves as likely to succeed without acquiring the relevant skills." | 1221 | M6, M28, C6, C10, C12, C13 |
| L29 | Corcoran and Karels [1985] | Identical players bet continuously in a Tullock lottery with free entry. The count is set by the lottery's exponent $r$, by the commitment assumed, or by a minimum bet. With a minimum bet $Z$, which the paper reads as a fixed entry fee, every entrant bets $Z$ and wins with probability $1/n$, so entry continues while $nZ \leq X$. That is the bound of M10 binding, because those who do not bet cannot win. Under hit-and-run entry several counts are stable equilibria, because incumbents choose among a range of pre-emptive bets, not because entrants differ. There is no wealth and no spread of entry costs. The record in `lit/corcoran_karels_1985` adds a closed form for the hardball count and lists misprints. | "In such a case, instead of a minimum bet requirement there is a fixed entry fee" | 232 | M6, M10, C5, C6, C13 |

| ID | Paper | What we rely on | Quote | Page | Model rows |
|---|---|---|---|---|---|
| L30 | Fernández and Galí [1999] | A continuum of agents, with ability and wealth uniform and independent, is matched to a continuum of schools, either by price or by a tournament that ranks a signal bought with a wasted, continuous expenditure. Utility is linear, so wealth matters only through a borrowing constraint; without one, both mechanisms give the efficient allocation. With one, higher ability relaxes the constraint in the tournament, which then gives more output than the market. Who spends nothing emits no signal and gets the lowest school. As the signal weighs ability more relative to spending, aggregate spending tends to zero, a limit in the direction of M28. No spread of wealth is signed, and there is no count of entrants and no share (record in `lit/fernandez_gali_1999`). *Unverified:* the working paper (NBER WP 5930, 1997) was read, and the *Review of Economic Studies* version (1999) is not checked. | "In a tournament, on the other hand, identical expenditures by non-identical agents do not lead to identical outcomes." | 28 | M28, C5, C12, C13 |
| L31 | Hassin and Haviv [1997] | Identical customers arrive at a queue and may buy preemptive priority at a fixed toll; those who do not buy are still served, later. Buying follows the crowd: the more others buy, the more one gains, so several thresholds can be equilibria. The paper's own framework states that a non-increasing best response, the case of our gain in the count (M4), has a single fixed point. There is no wealth, no ranking by merit and no fixed prize. The record shows that the stated bound on the number of pure equilibria, $\lfloor 1/(1-\rho) \rfloor$, is one short when $1/(1-\rho)$ is not an integer (`lit/hassin_haviv_1997`). *Unverified:* the authors' preprint was read, and the *Operations Research* version (1997) is not checked. | "In ATC case there is a single fixed point." | 4 | M4, M6, C3, C6, C13 |

| ID | Paper | What we rely on | Quote | Page | Model rows |
|---|---|---|---|---|---|
| L32 | Liu et al. [2022] | A continuum of applicants chooses a continuous, costly effort that multiplies skill into a score, and a school admits a fixed share by post-effort rank, with a reward set for each band of ranks. In equilibrium effort does not reorder applicants (rank preservation), so the design sets only what admission costs. With two groups whose score is scaled by their environment, the disadvantaged group's *access*, its admission probability, does not rise as the ranking becomes more decisive. Access is the construct nearest the entrants' share of M31, but the group is fixed by environment, not by a choice to pay, and there is no count of those who pay. The record (`lit/liu_garg_borgs_2022`) corrects two slips in the proof of the access result. *Unverified:* the arXiv version was read, and the AISTATS proceedings version (2022) is not checked. | "Access is the overall probability of admission of the disadvantaged group." | 8 | M31, C13 |

| ID | Paper | What we rely on | Quote | Page | Model rows |
|---|---|---|---|---|---|
| L33 | Hu et al. [2019] | Two groups can raise the feature a classifier reads, and raising it costs the disadvantaged group more; the paper derives that cost condition from the same outlay weighing more in its utility, which is the reading of M23. The learner sets one threshold for both groups, so it wrongly admits some of the advantaged group and wrongly rejects some of the disadvantaged. A subsidy to the disadvantaged group's costs can leave both groups worse off. There is a threshold rather than a fixed number of positions, the effort is continuous, and there is no distribution of wealth. The record (`lit/hu_immorlica_vaughan_2019`) notes a sign misprint in the appendix. *Unverified:* the arXiv version was read, and the FAT* proceedings version (2019) is not checked. | "Manipulating from a feature vector x to y is always at least as costly for a member of group B as it is for a member of group A." | 4 | M23, C13 |

Papers that `LITERATURE.tex` classifies from their abstracts only are not in the table, and no row rests on them.

# 5 Conclusions

Table 5 holds the conclusions, each resting on the model rows and on papers read in full, with two exceptions that the rows name: Morgan et al. [2022] were read in Sections I to IV (L20) and Tilly [1998] from one page (L19). A claim that something is new entered the table once every paper it waited on had been read. The last was Corcoran and Karels [1985], which settled C5, C6 and C13 (L29). The novelty reading is pass 7 of `TODO.md`. The two leads of the application search of 14 October 2026 closest to these claims, Fernández and Galí [1999] (L30) and Hassin and Haviv [1997] (L31), were then read in full; they change none of the claims. The search of 15 October 2026, on whether the question of Section 1.1 had been answered, found no paper that answers it; its two nearest neighbours, Liu et al. [2022] (L32) and Hu et al. [2019] (L33), were read in full and change none of the claims. C15 to C18 are implications for an institution that sets the fee, the prize or the weight. They say what follows from each choice, not which choice is better, since the model has no welfare objective. Figure 1 draws the conclusions as relations among the players and the objects of the measurement map.

Table 5: Conclusions that rest on the model rows and on the papers read.

| ID | Statement | Rests on |
|---|---|---|
| C1 | For a fixed prize awarded by rank, a pivot-spread of wealth moves the number of entrants weakly, in a direction set by one comparison, the marginal entrant against the pivot of the spread. Given score laws that do not depend on who enters, the rule needs only that the fee's utility cost falls with wealth. In the privilege contest of Schroyen and Treich the effect of a spread on effort turns on $u'''$; on the entry margin the sign needs no condition on $u'''$. Their paper was read in its working-paper version (L12). | M13, M23, M24, M26, L12 |

| ID | Statement | Rests on |
|---|---|---|
| C2 | The margin condition reads the wealth profile at two ranks only, the marginal entrant and the first outsider, and it holds for any change of profile, mean-preserving or not (M12). Where the marginal entrant becomes poorer and the first outsider richer, the signs of the change settle nothing, even for mean-preserving spreads, and the mean cannot stand in for the pivot (M13, M14). Costrell and Loury, in their two-job model, sign a spread by one marginal quantile fixed by technology (L9); here the margin is the count itself, which moves with the profile. | M12, M13, M14, L9 |
| C3 | Every pure-strategy equilibrium has the same number of entrants because the gain depends on that number alone and does not rise with it (M4, M26). When the gain depends on who enters, the count need not be pinned: for some laws in which the bought component falls (M27) or rises (M29) with wealth it is not, and our witness in the entry stage of Fullerton and McAfee, where profits depend on rivals' costs, has two counts (L5). In the priority queue of Hassin and Haviv buying follows the crowd, the opposite monotonicity, and several equilibria can coexist (L31). | M4, M6, M26, M27, M29, L5, L31 |
| C4 | Under complete information every pure-strategy equilibrium has $k^*$ entrants (M6), as in pure strategies with identical entrants (L2, L3). With heterogeneous costs the $k^*$ richest entering is always an equilibrium (M8), so who enters is settled without the devices that models with identical entrants use for that purpose: mixed entry (L2), private random delays (L3) and sequential arrival (L7). Who enters is pinned when the condition of M7 holds, and need not be otherwise. | M5, M6, M7, M8, L2, L3, L4, L7 |
| C5 | The sign rule of C1 is new among the papers read. None of the entry papers read signs the effect of a spread of wealth, or of a spread of entry costs, on the number of entrants at a fixed number of potential entrants. In Corcoran and Karels [1985], whose players are identical, the count is set by the lottery's exponent, the commitment assumed or a minimum bet (L29). The privilege contest of C1 signs effort, not entry (L12). Fernández and Galí, a rank-order allocation in which wealth matters, fix the wealth distribution and sign no spread (L30). | M13, L2, L3, L4, L5, L6, L7, L12, L23, L24, L29, L30 |
| C6 | Count invariance across pure-strategy equilibria with heterogeneous entry costs, for finitely many challengers, is new among the papers read. The entry papers read have identical entrants (Levin and Smith, Morgan, Orzen and Sefton, Fu, Jiao and Lu, Fu and Lu, Corcoran, Corcoran and Karels), a random count under private costs (Moreno and Wooders), or, in our witness for the entry stage of Fullerton and McAfee, a count that differs across equilibria. Mathews and Namoro fix the number of strategic entrants at two. In the continuum model of Fryer and Loury, with a binary investment and heterogeneous costs, the investment rate is unique and the investors are the agents below a cost threshold (L25), the continuum counterpart; with finitely many contestants the count is pinned (M6) while the identities need not be (the example in Appendix D, Information). In the continuum model of Coate and Loury the investment rate need not be unique, because the employer's standard responds to the rate it expects (L28); with the rule fixed, as here, that channel is absent. Under hit-and-run entry Corcoran and Karels find several stable counts, because incumbents choose among pre-emptive bets, not because entrants differ (L29). In Hassin and Haviv, identical customers buy priority at a fixed toll and several thresholds can be equilibria, because buying follows the crowd (L31). | M6, M7, L2, L3, L4, L5, L6, L7, L23, L24, L25, L28, L29, L31 |
| C7 | Beyond what C14 credits to Ryvkin and Drugov, two statements about the first entrant's gain are new among the papers read. With a uniform base component it never falls in the number of challengers, because the peak of the score gap sits at the top of the outsiders' support, where the kernel vanishes, which it cannot do in their tournament (M18). For a bought component of two values, a shift condition on the base component makes the score gap single-peaked (M19). The search of 13 October 2026 found one candidate for these statements, Fu and Lu [2009], whose investment is continuous and made by every contestant, so it is not a precedent (L26); the binary-investment models it also found, Fryer and Loury (L25) and Coate and Loury (L28), are continua with no number of challengers. | M18, M19, L8, L25, L26, L28 |
| C8 | Outside the band, two ranks suffice to sign the change in the count (M12, M14), a sharpening of Costrell and Loury [2004]. Read in their published version, they sign a spread by one marginal quantile fixed by technology, or by the shape of the weight on ability, and never by two ranks (L9). | M12, M14, L9 |
| C9 | With the fee, the prize and the wealth profile fixed, the number of entrants does not rise as the rule weighs the base component more, for every atomless law of the base component and every law of a non-negative bought component, when the incumbent is scored as an entrant. For an incumbent protected from the rule, whose law does not move with the weight, the argument does not apply (Appendix D). In the two effort models read the sign is the opposite (L20, L22): effort rises with the accuracy of the ranking until entry earns no more than staying out, while a fixed fee does not rise (Appendix D, "No effort margin"). | M28, L20, L22 |

| ID | Statement | Rests on |
|---|---|---|
| C10 | The share of the prize taken by the entrants is set by the count, so it moves weakly with the prize, the fee and the wealth profile in the direction the count does (M11 to M13), and it does not rise as the rule weighs the base component more. That a group with higher investment costs holds fewer positions is Fryer and Loury's result under laissez-faire, for identity groups in a continuum (L25); in Coate and Loury such a group does less well when employers set the standard (L28). The outcome is the opportunity hoarding of Tilly [1998], a bounded group keeping access to a valuable resource, read from one page (L19), but here no one hoards, and the fee and the wealth profile produce it. Laurison and Friedman [2016] find access to the traditional professions more closed than to management, which pays more. Read as shares, a larger prize alone does not lower the share, so the prize does not account for that pattern; a rule weighing the bought component more is one difference consistent with it (weakly, M31), though the weight is not measured. | M11, M12, M13, M28, M31, L17, L19, L25, L28 |
| C11 | An incumbent scored as an entrant, with the lowest cost, sits out while a challenger enters only as one of several equilibria with the same count, when $g(k^*) < \kappa_0$ and $\kappa_{k^*} \leq g(k^* - 1)$, and never when identities are pinned. | M6, M7, M30 |
| C12 | The sign of M28 on the entry margin, for a rule that weighs the base component more, is new among the papers read. The two noise-comparative entry models read have the opposite sign, more noise bringing more entrants, because effort rises as noise falls and competes away the rents of entry above staying out (L20; Drugov and Ryvkin's free-entry extension, L22). M28's sign is that of their intensive margin, effort falling as noise becomes more dispersed (L22), which we read as the same margin, since entry, which buys the bought component, is the investment. The contrast with the number of challengers, which has no universal sign (M18, M20, M22), is theirs across their two papers (L8, L22). The working paper of Fryer and Loury conjectures, without deriving it, that a blind rule rationing slots at random lowers investment in both groups, the direction of M28; the published version drops the conjecture (L25). In the pair-wise tournaments of Fryer and Loury (2005) a blind redistribution rule creates a pool of contestants who exert no effort and still compete, and the pool grows as the redistribution target becomes more aggressive, which the paper asserts and the record derives (L27): the same direction, for a redistribution target rather than a weight on the base component, and with a continuous effort rather than a fee. In Coate and Loury [1993] a noisy test reads a binary investment, and in their uniform example, at a fixed employer policy, their formulas imply that both investment rates fall as the test becomes less accurate, a derivation of ours that the paper does not state (L28): the direction of M28 again, for the accuracy of a test and a standard that employers set. In Fernández and Galí [1999], as the signal grows more sensitive to the agents' endowment relative to their spending, aggregate spending tends to zero (L30): the direction of M28 again, as a limit, for a continuous expenditure in a continuum. The sign that is new is C9's: a fixed prize awarded by rank, a fixed fee, the incumbent scored as an entrant, and every atomless law of the base component and every law of a non-negative bought component. | M18, M20, M22, M28, L8, L20, L22, L25, L27, L28, L30 |
| C13 | The share of the prize taken by those who enter, as a function of how many enter, and its comparative statics are new among the entry papers read. In Mathews and Namoro, non-strategic contestants fill the places that the strategic ones leave and can win, but they make no choice, the number of strategic contestants is fixed at two, and no share is analysed (L23). In every other entry model read, those who do not enter cannot win, so the share is not defined there; here outsiders compete on the base component. In the minimum-bet case of Corcoran and Karels every entrant wins with probability $1/n$ and those who do not bet cannot win (L29). Outside the entry literature, Fryer and Loury have a binary investment in which those who do not invest still compete, and the group with higher costs holds fewer slots (L25). Their agents form a continuum and slots are priced, so they do not derive the entrants' share as a function of how many enter, nor its comparative statics. In Coate and Loury [1993] those who do not invest can reach the better job and the assignment rate is split between investors and non-investors, but the number of places moves with the standard and the investors' share is not analysed (L28). In Fernández and Galí who spends nothing gets the lowest school (L30). In Hassin and Haviv those who do not buy priority are still served, but there is no prize to share (L31). The nearest construct is the access of Liu, Garg and Borgs, a group's admission probability for a fixed number of places, which does not rise as the ranking becomes more decisive; the group is fixed by environment and effort is continuous, so no share of those who pay is defined (L32). Hu, Immorlica and Vaughan give the errors of a threshold by group, with no fixed number of positions (L33). | M31, L2, L3, L4, L5, L6, L7, L20, L22, L23, L24, L25, L28, L29, L30, L31, L32, L33 |
| C14 | The first entrant's gain is $V\,\mathbb{E}[\varphi(M)]$, with $M$ the best rival's score (M1). This is the form of the marginal benefit of Ryvkin and Drugov [2020], the noise density at the best rival's shock, with the score gap $\varphi$ in place of the density. Their Karlin step, with the law of $M$ in place of $F^{k-1}$, gives M17, no interior minimum in the number of challengers when $\varphi$ is single-peaked. Their remark that with many players the comparative statics follow the upper tail is what M21 makes precise for the gain, in the class of M19. | M1, M17, M21, L8 |

| ID | Statement | Rests on |
|---|---|---|
| C15 | A fee equal for every challenger changes how many enter, not the order in which they enter. At every fee the order of entry is the order of wealth (M5, M23), and in the equilibrium in which the richest enter a lower fee adds entrants from the next-richest down (M8, M11). Fees that differ across challengers can reverse the order of costs (M33), so only a fee that varies with the challenger can bring a poorer challenger in ahead of a richer one. | M5, M8, M11, M23, M33 |
| C16 | The prize caps how many pay, and the cap sets a fee. At any fee at which $(k+1)\kappa_k$ exceeds $V$, at most $k$ challengers enter (M10, M33), so an institution that wants at most $k$ entrants needs only the prize and the cost of the fee to the $(k+1)$-th richest. | M10, M33 |
| C17 | Whether a compression of wealth raises participation is read at two ranks. A compression toward a pivot weakly raises the count when the marginal entrant stands at or below the pivot, and weakly lowers it when the first outsider stands at or above it (M12, M33); inside the band between them the direction is not determined (M14). That less inequality brings more participation has no general sign, since a spread that keeps the mean can lower the count (M13). | M12, M13, M14, M33 |
| C18 | A large share of positions among the holders of the bought component is not by itself evidence that it counts. When it adds nothing to the score, the entrants, the incumbent included, take $\frac{k+1}{Q+1}$ of the prize, their share of the players (M31). With a non-negative bought component the share is never below that benchmark, and a share off it means the two score laws differ (M33), so only a share above the benchmark shows that the bought component counts. This bears on how the opportunity hoarding of Tilly [1998] is read in data (L19). | M31, M33, L19 |

# A Proofs

**Notation.** Challenger $i$ scores $\sigma_i = \mu r_i + (1-\mu) s_i e_i$, with the base component $r_i$ and the bought component $s_i \geq 0$ drawn independently of each other and of wealth, $\mu$ the weight on the base component, and the fee $c > 0$. $F$ and $G$ are the score CDFs of an entrant and of an outsider, and $C$ is the incumbent's, each without atoms. With $m$ other entrants among the $Q$ challengers,

$$H_m = C F^m G^{Q-1-m}, \qquad K_m = C F^m G^{Q-2-m}, \qquad \varphi = G - F,$$

$$\Delta(m) = V\Big(\int H_m \, dF - \int H_m \, dG\Big), \qquad \kappa(w) = u(w) - u(w-c).$$

$H_m$ is the CDF of the best score facing a challenger, $M_m$ a random variable with that CDF, and $K_m$ the same CDF with one more challenger set aside, so that $H_{m+1} = K_m F$ and $H_m = K_m G$. Challengers are indexed $0, \ldots, Q-1$ in increasing order of cost, $\kappa_0 \leq \cdots \leq \kappa_{Q-1}$, which is decreasing order of wealth. Index $j$ enters when $\kappa_j \leq \Delta(j)$, and $k^*$ satisfies $\kappa_j \leq \Delta(j)$ for $j < k^*$ and $\Delta(k^*) < \kappa_{k^*}$ when $k^* < Q$. Past $Q-1$, $\Delta(m)$ means $\Delta(Q-1)$; so extended, $\Delta$ is non-increasing on every count by M4, and the results that use this cite M4. In the results on wealth profiles, rank $j$ has wealth $w_j$ with $w_0 \geq w_1 \geq \cdots$, and enters when $w_j > c$ and $\kappa(w_j) \leq \Delta(j)$, so a challenger whose wealth is at or below the fee does not enter. A prime marks the profile after a change and its count $k'^*$. When the number of challengers varies, $\Delta_Q(0)$ is the first entrant's gain among $Q$ challengers, with

$$D(Q) = \Delta_{Q+1}(0) - \Delta_Q(0), \qquad W_Q = C G^{Q-1}(1-G).$$

### Proof of M1

*Lean:* `EntryContestP1P2.two_max_sum`, `EntryContestP1P2.p1_representation`, `EntryContestP1P2.Delta_eq_expectation`.

For independent $X \sim A$ and $Y \sim B$ with $B$ atomless, $\Pr(X = Y) = 0$, so

$$\int B \, dA + \int A \, dB = \Pr(Y \leq X) + \Pr(X \leq Y) = 1.$$

With $(A, B) = (F, H_m)$ and $(A, B) = (G, H_m)$,

$$\int H_m \, dF = 1 - \int F \, dH_m, \qquad \int H_m \, dG = 1 - \int G \, dH_m,$$

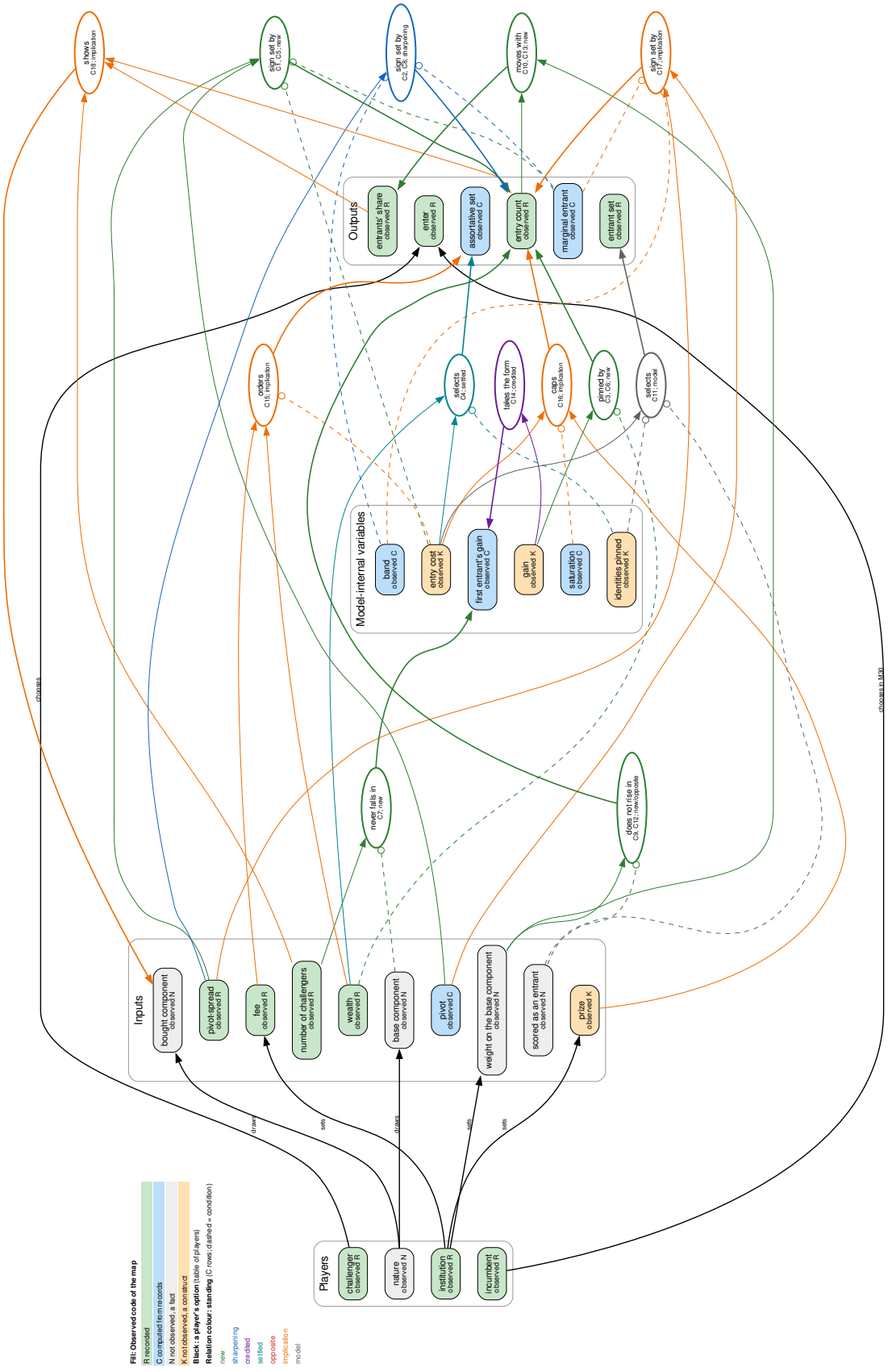



Figure 1: The conclusions as relations among the players and the objects of the measurement map. Boxes are rows of the map, in its four tables, filled by their Observed code. Black edges are a player's options from Table 2. Each ellipse is a relation, labelled with what it says, the conclusions it represents and their standing against the papers read; solid edges lead in from what changes and out to what responds, and dashed edges are conditions. The figure is drawn by `checks/ledger_literature.py` from its ledger (Appendix B).

so

$$\frac{\Delta(m)}{V} = \int (G-F)\, dH_m = \int \varphi\, dH_m = \mathbb{E}[\varphi(M_m)].$$

□

## Proof of M2

*Lean:* `EntryContestP1P2.p2_fosd`, `EntryContestP1P2.p2_strict_interval`, `EntryContestP1P2.Delta_nonneg_from_primitives`.

Let $X = \mu r + (1-\mu)s$ and $Y = \mu r$. For $\mu \le 1$ and $s \ge 0$, $X - Y = (1-\mu)s \ge 0$, so $\{X \le x\} \subseteq \{Y \le x\} \cup \{s < 0\}$ and

$$F(x) \le G(x) + \Pr(s < 0) = G(x), \qquad \Delta(m) = V \int \varphi\, dH_m \ge 0.$$

For $\mu < 1$,

$$\{s > 0\} \subseteq \{Y < X\} = \bigcup_{q \in \mathbb{Q}} \{Y \le q < X\},$$

so $\Pr(s > 0) > 0$ gives $q \in \mathbb{Q}$ with $\Pr(Y \le q < X) > 0$, and

$$G(q) \ge \Pr(X \le q) + \Pr(Y \le q < X) > F(q).$$

$G - F$ is right-continuous, so $G - F > 0$ on some $[q, u)$.

□

## Proof of M3

*Lean:* `EntryContestStep.three_max_sum_real`, `EntryContestStep.step_identity`, `EntryContestModel.Delta_step`.

Write $K = K_m$, so $H_{m+1} = KF$ and $H_m = KG$. For independent atomless draws $Z \sim K$, $X \sim A$, $Y \sim B$, exactly one is the largest,

$$\int AB\, dK + \int KB\, dA + \int KA\, dB = 1.$$

With $(A,B) = (F,F)$, $(G,G)$ and $(F,G)$,

$$\int KF\, dF = \tfrac{1}{2}\Big(1 - \int F^2 dK\Big), \quad \int KG\, dG = \tfrac{1}{2}\Big(1 - \int G^2 dK\Big), \quad \int KG\, dF + \int KF\, dG = 1 - \int FG\, dK,$$

so

$$\frac{\Delta(m+1) - \Delta(m)}{V} = \int KF\, dF - \int KF\, dG - \int KG\, dF + \int KG\, dG = -\tfrac{1}{2} \int (F-G)^2\, dK.$$

□

## Proof of M4

*Lean:* `EntryContestModel.Delta_step_nonpos`, `EntryContestModel.DeltaUpTo_step`, `EntryContestEq.Delta_step_strict`.

By M3 and $\varphi^2 \ge 0$,

$$\Delta(m+1) - \Delta(m) = -\tfrac{V}{2} \int \varphi^2\, dK_m \le 0 \quad (V \ge 0), \qquad < 0 \quad \Big(V > 0,\ \int \varphi^2\, dK_m > 0\Big).$$

Beyond $m = Q - 1$ the gain is set to $\Delta(Q-1)$, so $\Delta(m+1) \le \Delta(m)$ for every $m$.

□

## Proof of M5

*Lean:* `EntryContest.single_crossing`, `EntryContestModel.p5_threshold_from_primitives`.

For $i \le j$, by M4,

$$\kappa_i \le \kappa_j \le \Delta(j) \le \Delta(i).$$

□

## Proof of M6

*Lean:* `EntryContestEq.count_invariance_fin`, `EntryContestMM.exists_mem_ge`, `EntryContestMM.exists_not_mem_le`.

Let $S$ be an equilibrium and $k = |S|$. A $k$-set of indices has an element $i \ge k-1$, since $S \subseteq \{0,\dots,k-2\}$ gives $k \le k-1$, and omits some $j \le k$, since $\{0,\dots,k\} \subseteq S$ gives $k+1 \le k$. If $k > k^*$, then $k^* < Q$ and

$$\Delta(k^*) < \kappa_{k^*} \le \kappa_{k-1} \le \kappa_i \le \Delta(k-1) \le \Delta(k^*).$$

If $k < k^*$, then $j < Q$ and

$$\Delta(k) < \kappa_j \le \kappa_k \le \Delta(k).$$

Both are contradictions, so $k = k^*$.

□

## Proof of M7

*Lean:* `EntryContest.members_below_kstar`, `EntryContestEq.assortative_unique`, `EntryContestEq.assortative_only`, `EntryContestEq.assortative_only_model`.

A member $i$ of an equilibrium has $\kappa_i \le \Delta(k^*-1)$ by M6. If $i \ge k^*$,

$$\kappa_i \ge \kappa_{k^*} > \Delta(k^*-1),$$

so $i < k^*$. Then $S \subseteq \{0,\dots,k^*-1\}$ and $|S| = k^*$ give $S = \{0,\dots,k^*-1\}$, an equilibrium by M8. With $\Delta$ the model's gain, non-increasing by M4, nothing further is assumed.

□

## Proof of M8

*Lean:* `EntryContestEq.assortative_is_equilibrium`, `EntryContestEq.assortative_min_cost`, `EntryContestEq.prize_term_anonymous`, `EntryContestEq.payoff_gap_anonymous`.

(i) For $i < k^*$ and for $k^* \le j < Q$,

$$\kappa_i \le \kappa_{k^*-1} \le \Delta(k^*-1), \qquad \Delta(k^*) < \kappa_{k^*} \le \kappa_j.$$

(ii) Induct on $S$, adding its largest element $a$ last. With $T = S \setminus \{a\}$ and $T \subseteq \{0,\dots,a-1\}$, $|T| \le a$, so

$$\sum_{i<|T|+1} \kappa_i = \sum_{i<|T|} \kappa_i + \kappa_{|T|} \le \sum_{i\in T} \kappa_i + \kappa_a = \sum_{i\in S} \kappa_i.$$

(iii) With identical laws, the win probability of $i$ is $P^{\text{in}}(|S|) = \int CF^{|S|-1}G^{Q-|S|}\,dF$ for $i \in S$ and $P^{\text{out}}(|S|) = \int CF^{|S|}G^{Q-1-|S|}\,dG$ for $i \notin S$, so with $EU_i(S) = u(w_i) - \kappa_i \mathbf{1}\{i \in S\} + VP_i(S)$,

$$\sum_i EU_i(S) = \sum_i u(w_i) - \sum_{i\in S} \kappa_i + V\big(|S|P^{\text{in}}(|S|) + (Q-|S|)P^{\text{out}}(|S|)\big),$$

and the last term is the same for any two entrant sets of one size.

□

### Proof of M9

*Lean:* `EntryContestP6P7.integral_iid`, `EntryContestP6P7.integral_cdf_pow`, `EntryContestP6P7.p6_evaluation`, `EntryContestP6P7.p6_at_zero`, `EntryContestP6P7.tendsto_cdf_investorLaw`, `EntryContestP6P7.tendsto_cdf_nonInvestorLaw`, `EntryContestP6P7.p6_limit`.

For $X_0, \dots, X_k$ independent with atomless CDF $F$,

$$\int F^k \, dF = \Pr\Big(X_0 = \max_{i \le k} X_i\Big) = \frac{1}{k+1}.$$

At $\mu = 0$ the entrant scores $s$ and the outsider $0$, so $G = \mathbf{1}_{[0,\infty)}$ and $F(0) = \Pr(s \le 0) = 0$. With $C = F$,

$$\int H_m \, dF = \int F^{m+1} \, dF = \frac{1}{m+2}, \qquad \int H_m \, dG = H_m(0) = 0.$$

For $\mu \to 0$ let $X_\mu = \mu r + (1-\mu)s$ and $Y_\mu = \mu r$. For $s > 0$,

$$F_\mu(X_\mu) \to F_0(s), \qquad G_\mu(X_\mu) \to 1, \qquad F_\mu(Y_\mu) \to F_0(0) = 0,$$

with every integrand in $[0,1]$, so by dominated convergence

$$\int H_m \, dF_\mu \to \int F_0^{m+1} \, dF_0 = \frac{1}{m+2}, \qquad 0 \le \int H_m \, dG_\mu \le \mathbb{E}\big[F_\mu(Y_\mu)\big] \to 0.$$

□

### Proof of M10

*Lean:* `EntryContestP6P7.integral_cdf_pow`, `EntryContestP6P7.p7_Delta_le`, `EntryContestP6P7.p7_cap`, `EntryContestP6P7.p7_count`.

From $C \le 1$, $G \le 1$ and $\int H_m \, dG \ge 0$,

$$\Delta(m) \le V \int H_m \, dF \le V \int F^m \, dF = \frac{V}{m+1}.$$

For $\kappa \le \Delta(m)$,

$$(m+1)\,\kappa \le (m+1)\,\Delta(m) \le V, \qquad m+1 \le \frac{V}{\kappa} \quad (\kappa > 0),$$

and a member $i$ of an equilibrium $S$ has $\kappa_i \le \Delta(|S|-1)$, so $|S|\,\kappa_i \le V$.

□

### Proof of M11

*Lean:* `EntryContestSpread.count_ge_of_enters`, `EntryContestSpread.count_mono_gain`, `EntryContestSpread.count_mono_prize`, `EntryContestEq.Delta_mono_prize`, `EntryContestMM.Delta_scale`, `EntryContestSpread.count_antitone_fee`.

Rank $j$ enters when $c < w_j$ and $\kappa(w_j) \le \Delta(j)$, and $k^*$ is the length of the prefix of entrants. If every rank $j < k'$ enters under $(\kappa, \Delta)$ then $k' \le k^*$. Write $\Delta_V$ for the gain at prize $V$. By M1 and M2,

$$\Delta_V(j) = V\,\Delta_1(j), \quad \Delta_1(j) \ge 0 \quad (\mu \le 1,\ s \ge 0), \qquad V \le V' \ \Rightarrow\ \Delta_V(j) \le \Delta_{V'}(j),$$

so an entrant at $V$ enters at $V'$. For $c \le c'$ and $c' < w_j$,

$$c < w_j, \qquad 0 < w_j - c' \le w_j - c \ \Rightarrow\ u(w_j) - u(w_j - c) \le u(w_j) - u(w_j - c'),$$

so an entrant at $c'$ enters at $c$.

□

## Proof of M12

*Lean:* `EntryContestSpread.enters_down`, `EntryContestSpread.count_ge_of_enters`, `EntryContestSpread.count_le_of_fails`, `EntryContestSpread.margin_rise`, `EntryContestSpread.margin_fall`.

(i) The marginal entrant has $c < w_{k^*-1} \le w'_{k^*-1}$, so

$$\kappa(w'_{k^*-1}) \le \kappa(w_{k^*-1}) \le \Delta(k^*-1),$$

and for $i \le k^*-1$, with $w'_i \ge w'_{k^*-1} > c$,

$$\kappa(w'_i) \le \kappa(w'_{k^*-1}) \le \Delta(k^*-1) \le \Delta(i).$$

Every $i < k^*$ enters after the change, so $k^* \le k'^*$. (ii) For $k^* < Q$, rank $k^*$ fails before the change. If $w'_{k^*} > c$, then $c < w'_{k^*} \le w_{k^*}$ and

$$\kappa(w'_{k^*}) \ge \kappa(w_{k^*}) > \Delta(k^*),$$

so rank $k^*$ fails after the change and $k'^* \le k^*$.

□

## Proof of M13

*Lean:* `EntryContestSpread.pivot_rise`, `EntryContestSpread.pivot_fall`, `EntryContestSpread.spread_isPivot`, `EntryContestMM.mm2a`, `EntryContestSpread.mps_lowers_count`, `EntryContestSpread.mps_lowers_witness_counts`, `EntryContestSpread.mps_lowers_count_positive`.

$T$ is non-decreasing, so $w'_j = T(w_j)$ is sorted, and

$$x_0 \le w_{k^*-1} \;\Rightarrow\; w_{k^*-1} \le T(w_{k^*-1}), \qquad w_{k^*} \le w_{k^*-1} \le x_0 \;\Rightarrow\; T(w_{k^*}) \le w_{k^*},$$

so M12 applies. For $T(w) = x_0 + \lambda(w - x_0)$ with $\lambda \ge 1$,

$$T(w) - w = (\lambda - 1)(w - x_0), \qquad \frac{1}{Q}\sum_j T(w_j) = \bar{w} + \lambda\left(\frac{1}{Q}\sum_j w_j - \bar{w}\right) = \bar{w} \quad (x_0 = \bar{w}).$$

With $c = 1$, $\kappa(w) = \frac{1}{w-1} - \frac{1}{w}$ and $\Delta(m) = \frac{1}{m+2}$,

$$w = \left(\tfrac{29}{10}, \tfrac{14}{5}, 2\right), \quad w' = \left(\tfrac{18}{5}, \tfrac{11}{5}, \tfrac{19}{10}\right), \quad \textstyle\sum_j w_j = \sum_j w'_j, \quad w'_2 \le w_2, \quad w'_2 + w'_1 \le w_2 + w_1,$$

$$\kappa(w) = \left(\tfrac{100}{551}, \tfrac{25}{126}, \tfrac{1}{2}\right), \quad \kappa(w') = \left(\tfrac{25}{234}, \tfrac{25}{66}, \tfrac{100}{171}\right), \quad \tfrac{1}{3}\textstyle\sum_j w_j = \tfrac{77}{30} < w_1,$$

so the counts are 2 and 1. They hold for any $\Delta$ with

$$\tfrac{100}{551} \le \Delta(0), \quad \tfrac{25}{126} \le \Delta(1) < \tfrac{25}{66}, \quad \Delta(2) < \tfrac{1}{2},$$

strict at $\Delta(m) = \frac{1}{m+2}$, so by M9 they hold for every small $\mu > 0$ with $r$ and $s$ uniform.

□

## Proof of M14

*Lean:* `EntryContestSpread.branch_of_not_band`, `EntryContestSpread.not_band_of_single_crossing`, `EntryContestSpread.pivot_single_crossing`, `EntryContestSpread.mps_band_signs`, `EntryContestSpread.mps_witness_counts`, `EntryContestSpread.mps_band_both_directions`, `EntryContestSpread.mps_band_both_directions_positive`.

Outside the band $d_{k^*-1} \ge 0$ or $d_{k^*} \le 0$, and M12 signs the change. If

$$i \le j,\ d_j > 0 \;\Rightarrow\; d_i \ge 0,$$

then $d_{k^*} > 0$ gives $d_{k^*-1} \geq 0$. A pivot-spread has this property, since $d_j > 0$ gives $w_j > x_0$ and then $w_i \geq w_j > x_0$. With $\kappa$ and $\Delta$ as in M13, the profiles

$$w = \left(4, \tfrac{14}{5}, \tfrac{11}{5}, 2\right), \quad w' = \left(\tfrac{41}{10}, \tfrac{27}{10}, \tfrac{13}{5}, \tfrac{8}{5}\right), \quad w'' = \left(\tfrac{23}{5}, \tfrac{9}{4}, \tfrac{9}{4}, \tfrac{19}{10}\right)$$

all sum to 11, with partial sums from the poorest up

$$\left(2, \tfrac{21}{5}, 7\right), \quad \left(\tfrac{8}{5}, \tfrac{21}{5}, \tfrac{69}{10}\right), \quad \left(\tfrac{19}{10}, \tfrac{83}{20}, \tfrac{32}{5}\right),$$

and

$$\kappa(w) = \left(\tfrac{1}{12}, \tfrac{25}{126}, \tfrac{25}{66}, \tfrac{1}{2}\right), \quad \kappa(w') = \left(\tfrac{100}{1271}, \tfrac{100}{459}, \tfrac{25}{104}, \tfrac{25}{24}\right), \quad \kappa(w'') = \left(\tfrac{25}{414}, \tfrac{16}{45}, \tfrac{16}{45}, \tfrac{100}{171}\right)$$

against $\Delta = \left(\tfrac{1}{2}, \tfrac{1}{3}, \tfrac{1}{4}, \tfrac{1}{5}\right)$. The counts are 2, 3 and 1, and $d'$ and $d''$ both have signs $(+,-,+,-)$. Every inequality is strict, so by M9 the counts hold for every small $\mu > 0$.

□

## Proof of M15

*Lean:*
`EntryContestKappa.exit_below_pivot`, `EntryContestKappa.enters_downward`, `EntryContestKappa.p9_strict`, `EntryContestKappa.p9_monotone`, `EntryContestKappa.p9_strict_model`.

Let $j$ be an entrant at $\lambda = 1$ with $w_j < x_0$. Since $\kappa(v) \to \infty$ as $v \downarrow c$, some $u > c$ has $\kappa > \Delta(j)$ on $(c,u)$, and

$$\lambda > \bar{\lambda} = \max\left(1, \frac{x_0 - u}{x_0 - w_j}\right) \;\Rightarrow\; x_0 + \lambda(w_j - x_0) < u,$$

so rank $j$ fails at $\lambda$. By the prefix property no rank from $j$ on enters, and

$$k^*(\lambda) \leq j < j+1 \leq k^*(1).$$

For $1 \leq \lambda_1 \leq \lambda_2$, ranks with $w_i \geq x_0$ have $i < j$ and enter at $\lambda_1$, while ranks with $w_i < x_0$ have $x_0 + \lambda_2(w_i - x_0) \leq x_0 + \lambda_1(w_i - x_0)$, so every entrant at $\lambda_2$ enters at $\lambda_1$.

□

## Proof of M16

*Lean:*
`EntryContestPMU.weight_eq`, `EntryContestPMU.pmu_identity`, `EntryContestPMU.pmu_sign_iff`, `EntryContestPMU.kernel_rises`, `EntryContestPMU.kernel_falls`, `EntryContestPMU.kernel_max`.

With no other entrant, $H_0 = CG^{Q-1}$ among $Q$ challengers and $CG^Q$ among $Q+1$, and $CG^{Q-1} - CG^Q = W_Q$, so

$$\frac{D(Q)}{V} = \int \left(CG^Q - CG^{Q-1}\right) dF - \int \left(CG^Q - CG^{Q-1}\right) dG = -\left(\int W_Q \, dF - \int W_Q \, dG\right).$$

For the kernel,

$$\frac{d}{dg} g^n (1-g) = g^{n-1}\left(n - (n+1)g\right) \quad (n \geq 1), \qquad \frac{d}{dg}(1-g) = -1 \quad (n = 0).$$

□

### Proof of M17

*Lean:* `EntryContestPMU.weight_shift`, `EntryContestPMU.pmu_single_crossing`, `EntryContestPMU.pmu_step_nonpos_persists`, `EntryContestPMU.quasiconcave_of_steps`, `EntryContestPMU.pmu_quasiconcave`.

Let $k = Q_2 - Q_1$, $\rho = G(x_0)^k$ and $\nu = dF - dG$. By M16 and $W_{Q_2} = G^k W_{Q_1}$,

$$D(Q_2) - \rho\, D(Q_1) = -V \int \left(G^k - \rho\right) W_{Q_1}\, d\nu.$$

$G$ is non-decreasing and $W_{Q_1} \geq 0$, so

$$\left(G^k - \rho\right) W_{Q_1} \leq 0,\ \nu \leq 0 \text{ on } (-\infty, x_0], \qquad \left(G^k - \rho\right) W_{Q_1} \geq 0,\ \nu \geq 0 \text{ on } (x_0, \infty),$$

and $D(Q_2) \leq \rho\, D(Q_1)$. Then $D(i) \leq 0$ gives $D(j) \leq 0$ for $j \geq i$. If $D(i) \leq 0$ for some $Q_1 \leq i < Q_2$, then $\Delta_{Q_3}(0) \leq \Delta_{Q_2}(0)$, and otherwise $\Delta_{Q_1}(0) \leq \Delta_{Q_2}(0)$.

□

### Proof of M18

*Lean:* `EntryContestPMU.uniform_left`, `EntryContestPMU.uniform_right`, `EntryContestPMU.pmu_quasiconcave_uniform`, `EntryContestRefute.pmu_step_nonneg_uniform`, `EntryContestRefute.Delta_mono_uniform`.

$G$ is the law of $\mu r$, so $G = 1$ on $[\mu, \infty)$, $dG = 0$ on $(\mu, \infty)$ and $W_Q = 0$ there. For $B \subseteq (-\infty, \mu]$ and $y \geq 0$, with $\tau = (1-\mu)y/\mu$ and $\lambda$ Lebesgue measure,

$$\Pr\left(\mu r + (1-\mu)y \in B\right) = \lambda\left([\tau, 1] \cap \tfrac{B}{\mu}\right) \leq \lambda\left([0,1] \cap \tfrac{B}{\mu}\right) = \Pr(\mu r \in B),$$

so $dF \leq dG$ on $(-\infty, \mu]$ and M17 applies at $x_0 = \mu$. By M16,

$$\frac{D(Q)}{V} = \int_{(-\infty, \mu]} W_Q\, d(G - F) \geq 0.$$

□

### Proof of M19

*Lean:* `EntryContestShift.left_of_lift`, `EntryContestShift.right_of_lift`, `EntryContestShift.quasiconcave_of_shiftMono`, `EntryContestShift.shiftMono_of_density`, `EntryContestShift.lawA_shiftMono`, `EntryContestShift.witness_quasiconcave`.

Let $t = 1 - \mu$ and $G_t$ be the law of $\mu r + t$, so $F = pG + (1-p)G_t$. From $\Pr(r \leq 0) = 0$, $G_t$ has no mass on $(-\infty, t]$, where $dF = p\, dG \leq dG$. For $B \subseteq (t, \infty)$, with $c = t/\mu$ and $B/\mu \subseteq (c, \infty)$,

$$G(B) = \Pr\left(r \in \tfrac{B}{\mu}\right) \leq \Pr\left(r \in \tfrac{B}{\mu} - c\right) = G_t(B),$$

so $dF \geq dG$ on $(t, \infty)$ and M17 applies at $x_0 = t$. For a density $g$ non-increasing on $(0, \infty)$ and $B \subseteq (c, \infty)$,

$$\int_B g(x)\, dx \leq \int_B g(x - c)\, dx = \int_{B-c} g(x)\, dx,$$

and the smaller of two uniform draws has $\Pr(r \leq x) = 1 - (1-x)^2$ and density $2(1-x)$ on $[0,1]$.

□

## Proof of M20

*Lean:* `EntryContestFall.stepQ_one`, `EntryContestFall.stepQ_two`, `EntryContestFall.rise_then_fall`, `EntryContestFall.fall_from_one`.

Here $G(x) = 1 - \left(1 - \frac{4x}{3}\right)^2$ on $[0, \frac{3}{4}]$, $F(x) = pG(x) + (1-p)G(x - \frac{1}{4})$, $W_1 = F(1-G)$ and $W_2 = FG(1-G)$. By M16, integrating the piecewise polynomials on $[0, \frac{1}{4}]$, $[\frac{1}{4}, \frac{3}{4}]$ and $[\frac{3}{4}, 1]$,

$$\frac{D(1)}{V} = -\frac{11p^2}{162} + \frac{1901p}{21870} - \frac{208}{10935}, \qquad \frac{D(2)}{V} = -\frac{4933p^2}{393660} + \frac{77219p}{2755620} - \frac{10672}{688905},$$

so

$$p = \tfrac{1}{2}: \ \frac{D(1)}{V} = \frac{653}{87480}, \ \frac{D(2)}{V} = -\frac{10169}{2204496}, \qquad p = \tfrac{1}{4}: \ \frac{D(1)}{V} = -\frac{179}{116640}.$$

□

## Proof of M21

*Lean:* `EntryContestMustFall.must_fall_general`, `EntryContestMustFall.shift_dominates`, `EntryContestMustFall.stepQ_bern`, `EntryContestMustFall.must_fall`, `EntryContestMustFall.witness_must_fall`, `EntryContestMustFall.unif_shiftMono`, `EntryContestMustFall.uniform_never_gains`.

With $F = pG + (1-p)G_t$ and $W_Q = G^{Q-1}W_1$, M16 gives

$$D(Q) = -V(1-p)\Big(\int W_Q\, dG_t - \int W_Q\, dG\Big).$$

$G_t$ has no mass on $(-\infty, t]$, $dG_t \geq dG$ on $(t, \infty)$ by M19, and $W_1 \geq \eta = C(a)\big(1 - G(b)\big) > 0$ on $(a, b]$. Splitting at $t$ and $a$,

$$\int W_Q\, dG_t - \int W_Q\, dG \ \geq \ G(a)^{Q-1}A - G(t)^{Q-1}B,$$

$$A = \int_{(a,\infty)} W_1\, d(G_t - G) \geq \eta\,(G_t - G)\big((a,b]\big) > 0, \qquad B = \int_{(-\infty,t]} W_1\, dG \geq 0.$$

With $r_0 = G(t)/G(a) < 1$ and $r_0^{Q_0-1}(B+1) < A$, the bound is positive for $Q \geq Q_0$. For M20's base law, $t = \frac{1}{4}$ and

$$G\big((\tfrac{1}{2}, \tfrac{5}{8}]\big) = \tfrac{35}{36} - \tfrac{8}{9} = \tfrac{1}{12} < \tfrac{7}{36} = \tfrac{3}{4} - \tfrac{5}{9} = G_t\big((\tfrac{1}{2}, \tfrac{5}{8}]\big).$$

A uniform base component is shift-monotone, and M18 gives $D(Q) \geq 0$, so it never meets the condition.

□

## Proof of M22

*Lean:* `EntryContestPMUWitness.pairA_fosd`, `EntryContestPMUWitness.pairB_fosd`, `EntryContestPMUWitness.pairA_step_one`, `EntryContestPMUWitness.pairA_step_two`, `EntryContestPMUWitness.pairB_step`, `EntryContestPMUWitness.fosd_does_not_sign`.

Pair one, with $y = 1 - x$, $W_1 = x(1-x)^2$, $W_2 = x\big(1 - (1-x)^2\big)(1-x)^2$ and $dG = 2(1-x)\,dx$, gives

$$\int W_1\, dF = \tfrac{1}{12}, \quad \int W_1\, dG = \tfrac{1}{10}, \quad \int W_2\, dF = \int_0^1 (y^2 - y^3 - y^4 + y^5)\, dy = \tfrac{1}{20}, \quad \int W_2\, dG = \tfrac{11}{210},$$

$$\frac{D(1)}{V} = \tfrac{1}{10} - \tfrac{1}{12} = \tfrac{1}{60}, \qquad \frac{D(2)}{V} = \tfrac{11}{210} - \tfrac{1}{20} = \tfrac{1}{420}.$$

Pair two, with $W_Q = x^{Q+1} - x^{Q+2}$ and $dF = 2x\,dx$, gives

$$\int W_Q\, dF = \frac{2}{(Q+3)(Q+4)}, \qquad \int W_Q\, dG = \frac{1}{(Q+2)(Q+3)}, \qquad \frac{D(Q)}{V} = -\frac{2(Q+2) - (Q+4)}{(Q+2)(Q+3)(Q+4)}.$$

□

## Proof of M23

*Lean:* `EntryContestKappa.burden_strictAnti_of_strictConcave`, `EntryContestBM.burden_strictAnti_of_deriv`, `EntryContestBM.not_burden_of_convex_stretch`, `EntryContestKappa.log_burden_strictAnti`, `EntryContestKappa.crra_strictConcave`, `EntryContestKappa.crra_burden_strictAnti`.

For $u$ strictly concave and $c < a < b$, secant slopes fall as either end moves right, and $a-c < a < b$, $a-c < b-c < b$ give

$$\frac{u(a)-u(a-c)}{c} > \frac{u(b)-u(a-c)}{b-a+c} > \frac{u(b)-u(b-c)}{c},$$

so $\kappa(a) > \kappa(b)$. With $u$ differentiable,

$$\kappa'(w) = u'(w) - u'(w-c) < 0 \quad (w > c).$$

If $u$ is convex on $[a,b]$ with $b-a > c$, then $a < b-c$ and

$$\kappa(a+c) = u(a+c) - u(a) \le u(b) - u(b-c) = \kappa(b).$$

Log and $x^{1-\gamma}/(1-\gamma)$ have $u'' < 0$ on $(0,\infty)$.

□

## Proof of M24

*Lean:* `EntryContestBM.rampU_burden`, `EntryContestBM.rampU_kappa_diverges`, `EntryContestBM.rampU_strictConvex`, `EntryContestBM.rampU_not_concave`, `EntryContestBM.admissibleParams_open`, `EntryContestBM.rampU_admissible`, `EntryContestBM.admissible_width`, `EntryContestBM.bm_strictly_weaker`.

Let $\rho$ rise linearly from 0 at $x_0$ to 1 at $x_0+\sigma$, and

$$u(x) = \log x + x + \eta \int_0^x \rho(t)\,dt, \qquad u'(x) = \tfrac{1}{x} + 1 + \eta\rho(x) > 0.$$

On $[x_0, x_0+\sigma]$, $u'' = -x^{-2} + \eta/\sigma > 0$ when $\sigma < \eta x_0^2$, so $u$ is strictly convex there and not concave. For $w > c$, $\rho(w) - \rho(w-c) \in [0,1]$ and vanishes unless $x_0 < w < x_0 + \sigma + c$, where

$$\tfrac{1}{w-c} - \tfrac{1}{w} = \tfrac{c}{w(w-c)} > \tfrac{c}{(x_0+\sigma+c)(x_0+\sigma)} > \eta,$$

so $u'(w) < u'(w-c)$ and M23 applies. $u(x) \to -\infty$ as $x \downarrow 0$, so M25 gives the divergence. The conditions

$$x_0 > 0, \quad \sigma > 0, \quad \sigma < \eta x_0^2, \quad \eta(x_0+\sigma+c)(x_0+\sigma) < c$$

are strict inequalities in continuous functions, so they define an open set, and for $\sigma < c$ an $\eta$ between $\sigma/x_0^2$ and $c/\big((x_0+\sigma+c)(x_0+\sigma)\big)$ exists once $x_0$ is large, since $\sigma(x_0+\sigma+c)(x_0+\sigma)/x_0^2 \to \sigma < c$.

□

## Proof of M25

*Lean:* `EntryContestKappa.kappa_diverges_iff`, `EntryContestKappa.not_diverges_of_continuousWithinAt_zero`, `EntryContestKappa.log_kappa_diverges`, `EntryContestKappa.crra_kappa_diverges`, `EntryContestKappa.crra_kappa_not_diverges`.

$u(w) \to u(c)$ as $w \downarrow c$, so

$$\kappa(w) = u(w) - u(w-c) \to \infty \iff u(w-c) \to -\infty \iff u(x) \to -\infty \ (x \downarrow 0).$$

$\log x \to -\infty$, and

$$\frac{x^{1-\gamma}}{1-\gamma} \to -\infty \quad (\gamma > 1), \qquad \frac{x^{1-\gamma}}{1-\gamma} \to 0 \quad (\gamma < 1).$$

□

## Proof of M26

*Lean:* `EntryContestAnon.profile_mem`, `EntryContestAnon.profile_not_mem`, `EntryContestAnon.cdf_profile`, `EntryContestAnon.gain_anonymous`.

Challenger $i$ wins with probability $\int C\prod_{j\neq i}L_j\,dL_i$, where $L_j$ is $F$ for an entrant and $G$ for an outsider. With $|T|=m$ and $i\notin T$,

$$\frac{\mathrm{gain}_i(T)}{V}=\int CF^mG^{Q-1-m}\,dF-\int CF^mG^{Q-1-m}\,dG=\int H_m\,dF-\int H_m\,dG=\frac{\Delta(m)}{V}.$$

□

## Proof of M27

*Lean:* `EntryContestAnon.winProb_zero_entrant`, `EntryContestAnon.winProb_zero_outsider`, `EntryContestAnon.gain_zero`, `EntryContestAnon.kappa3_values`, `EntryContestAnon.gains3`, `EntryContestAnon.anon_witness_zero`, `EntryContestAnon.gain_tendsto`, `EntryContestAnon.anon_witness_positive`, `EntryContestAnon.anon_witness_exists`.

Let $r$ be uniform on $[0,1]$, entrant $j$'s $s_j$ the largest of $n_j$ uniform draws, the incumbent's the largest of one, and $\mu=0$. An outsider scores 0, and among the $1+\sum_{j\in S}n_j$ draws each is equally likely to be the largest, so

$$\Pr(i\text{ wins})=\frac{n_i}{1+\sum_{j\in S}n_j},\qquad \mathrm{gain}_i(T)=\frac{V\,n_i}{1+n_i+\sum_{j\in T}n_j}.$$

With $V=1$, $c=1$, $u(x)=-1/x$, $w=(\frac{11}{4},\frac{9}{4},2)$ and $n=(1,2,3)$,

$$\kappa=\left(\tfrac{16}{77},\tfrac{16}{45},\tfrac{1}{2}\right),\qquad \mathrm{gain}_2(\emptyset)=\tfrac{3}{4},\ \mathrm{gain}_0(\{2\})=\tfrac{1}{5},\ \mathrm{gain}_1(\{2\})=\tfrac{1}{3},$$

$$\mathrm{gain}_0(\{1\})=\tfrac{1}{4},\ \mathrm{gain}_1(\{0\})=\tfrac{1}{2},\ \mathrm{gain}_2(\{0,1\})=\tfrac{3}{7},$$

so $\{2\}$ and $\{0,1\}$ are equilibria. Every inequality is strict and each gain is continuous in $\mu$ at 0, so both remain equilibria for small $\mu>0$.

□

## Proof of M28

*Lean:* `EntryContestMu.Delta_normalise`, `EntryContestMu.cdf_payerLaw_antitone`, `EntryContestMu.first_integral`, `EntryContestMu.Delta_payer_mono`, `EntryContestMu.Delta_antitone_mu`, `EntryContestMu.kstar_antitone_mu`.

For $\mu>0$ put $t=(1-\mu)/\mu$, $F_t$ the law of $r+ts$ and $G$ the law of $r$. Dividing every score by $\mu$ changes no rank, so with $C=F$ and $n=Q-1-m$,

$$\Delta(m;\mu)=V\Big(\int F_t^{m+1}G^n\,dF_t-\int F_t^{m+1}G^n\,dG\Big).$$

For $s\geq 0$ and $t\leq t'$, $r+t's\geq r+ts$, so $F_{t'}(x)\leq F_t(x)$ for every $x$, and the second integral is non-increasing in $t$. For the first, with $n\geq 1$,

$$\int F_t^{m+1}G^n\,dF_t=\frac{1}{m+2}\int G^n\,d\big(F_t^{m+2}\big)=\frac{1}{m+2}\Big(1-\int F_t^{m+2}\,d\big(G^n\big)\Big),$$

by $\int gF^k\,dF=\frac{1}{k+1}\int g\,d(F^{k+1})$ and $\int A\,dB+\int B\,dA=1$ for atomless $B$; the last integral is non-increasing in $t$. For $n=0$ the first integral is $1/(m+2)$. So $\Delta(m;\mu)$ is non-decreasing in $t$, and $t$ is decreasing in $\mu$. With $\kappa$ fixed, if $k'^*<k^*$ then

$$\Delta(k'^*;\mu')<\kappa_{k'^*}\leq\Delta(k'^*;\mu)\leq\Delta(k'^*;\mu'),$$

so $k^*(\mu)\leq k^*(\mu')$ for $\mu'\leq\mu$.

□

## Proof of M29

*Lean:* `EntryContestAligned.kappaA_values`, `EntryContestAligned.ability_with_wealth`, `EntryContestAligned.costs_ordered`, `EntryContestAligned.gainsA`, `EntryContestAligned.aligned_witness_zero`, `EntryContestAligned.aligned_witness_positive`, `EntryContestAligned.aligned_witness_exists`, `EntryContestAligned.control_window`.

Let $V = 1$, $r$ be uniform on $[0,1]$, $s_j$ the largest of $n_j$ uniform draws with $n = (12,10,9)$, the incumbent's the largest of one, and $\mu = 0$. An outsider scores 0 and never wins, and for $i \in S$,

$$\mathrm{gain}_i(S \setminus \{i\}) = \frac{n_i}{n_i + 1 + \sum_{j \in S \setminus \{i\}} n_j},$$

so

$$\mathrm{gain}_0(\emptyset) = \tfrac{12}{13},\ \mathrm{gain}_0(\{1,2\}) = \tfrac{3}{8},\ \mathrm{gain}_1(\{0\}) = \tfrac{10}{23},\ \mathrm{gain}_1(\{2\}) = \tfrac{1}{2},\ \mathrm{gain}_2(\{0\}) = \tfrac{9}{22},\ \mathrm{gain}_2(\{1\}) = \tfrac{9}{20}.$$

With $u(x) = -1/x$, $c = 1$ and $w = (\frac{43}{20}, \frac{52}{25}, \frac{83}{40})$, $\kappa = (\frac{400}{989}, \frac{625}{1404}, \frac{1600}{3569})$, increasing in the index, and

$$\tfrac{3}{8} < \kappa_0 \le \tfrac{12}{13}, \qquad \tfrac{10}{23} < \kappa_1 \le \tfrac{1}{2}, \qquad \tfrac{9}{22} < \kappa_2 \le \tfrac{9}{20},$$

so $\{0\}$ and $\{1,2\}$ are equilibria. Every inequality is strict and each gain is continuous in $\mu$ at 0, so both remain equilibria for small $\mu > 0$. Raising $\kappa_2$ to $\frac{23}{50} > \frac{9}{20}$ removes $\{1,2\}$.

□

## Proof of M30

*Lean:* `EntryContestInc.Delta_incumbent_in_eq`, `EntryContestInc.symGain_step`, `EntryContestInc.Delta_incumbent_in_incGain`, `EntryContestInc.incGain_top_step`, `EntryContestInc.incGain_antitone`, `EntryContestInc.incumbent_out_iff`, `EntryContestInc.incumbent_out_iff_model`, `EntryContestInc.incumbent_out_not_unique`, `EntryContestInc.incumbent_assortative_model`, `EntryContestInc.incumbent_in_of_pinned`, `EntryContestInc.incumbent_in_of_pinned_model`, `EntryContestInc.incumbent_count_model`, `EntryContestInc.band_witness`, `EntryContestInc.control_no_band`.

Players $0,\dots,Q$, costs $\kappa_0 \le \cdots \le \kappa_Q$, gain $g(m)$ with $m$ others in. For $m+2 \le Q$,

$$F \cdot F^m G^{Q-1-m} = G \cdot F^{m+1} G^{Q-2-m},$$

so $\Delta(m)|_{C=F} = g(m+1)$, and $g(m+1) \le g(m)$ by M4. At $m = Q-1$ both sides are $g(Q)$, and with $K = F^{Q-1}$ the step identity of M3 gives

$$g(Q) - g(Q-1) = -\tfrac{V}{2} \int \varphi^2 \, dK \le 0 \quad (Q \ge 2),$$

while for $Q = 1$, $\int F\,dF = \int G\,dG = \frac{1}{2}$ and $\int G\,dF + \int F\,dG = 1$ give $g(1) = g(0)$. So $g$ is non-increasing and M6 gives the count. Let $S$ be an equilibrium with $0 \notin S \neq \emptyset$. By M6, $|S| = k^* \ge 1$; $S \subseteq \{1,\dots,Q\}$ gives $k^* \le Q$; $0 \notin S$ gives $g(k^*) < \kappa_0$; and $|\{1,\dots,k^*-1\}| = k^* - 1 < |S|$ gives $i \in S$ with $i \ge k^*$, so

$$\kappa_{k^*} \le \kappa_i \le g(k^* - 1).$$

Conversely, let $S = \{1,\dots,k^*\}$. For $i \in S$, $\kappa_i \le \kappa_{k^*} \le g(k^*-1) = g(|S|-1)$, and for $j \notin S$, $g(|S|) = g(k^*) < \kappa_0 \le \kappa_j$. If $g(k^*-1) < \kappa_{k^*}$ the condition fails, so $0 \in S$ for every non-empty equilibrium $S$; $\{0,\dots,k^*-1\}$ is an equilibrium by M8. For $\kappa = (2,5,8)$ and $g = (12,10,1)$, $k^* = 2$, $g(2) = 1 < 2$, $\kappa_2 = 8 \le 10 = g(1)$, and $\{1,2\}$ is an equilibrium; with $\kappa_2 = 11 > 10$ no equilibrium omits 0.

□

## Proof of M31

*Lean:* `EntryContestShare.payerShare_full`, `EntryContestShare.payerShare_eq_one_sub`, `EntryContestShare.payerShare_monotone`, `EntryContestShare.payerShare_at_equal`, `EntryContestShare.payerShare_normalise`, `EntryContestShare.payerShare_antitone_mu`, `EntryContestShare.payerShare_kstar_antitone_mu`, `EntryContestShare.control_share_needs_nonneg`.

The best of $k+1$ entrants has law $F^{k+1}$ and the best of $n = Q-k$ outsiders $G^n$, so $\xi(k) = \Pr(\text{best entrant} > \text{best outsider})$, and $\xi(Q) = 1$. For $n \geq 1$, by $\int A\,dB + \int B\,dA = 1$ for atomless $B$,

$$\xi(k) = 1 - \int F^{k+1}\,d(G^n),$$

$$\xi(k+1) - \xi(k) = \int F^{k+1}\,d(G^n) - \int F^{k+2}\,d(G^{n-1}) \geq 0,$$

since $F^{k+2} \leq F^{k+1}$ and $\int F^{k+1}\,d(G^{n-1}) \leq \int F^{k+1}\,d(G^n)$ by the same identity and $G^n \leq G^{n-1}$. For $F = G$,

$$\xi(k) = (k+1)\int F^{Q-k}F^k\,dF = \frac{k+1}{Q+1}.$$

For $\mu > 0$, dividing every score by $\mu$ changes no rank, so $\xi$ is computed from $F_t$ and $G$ with $t = (1-\mu)/\mu$; $F_{t'} \leq F_t$ for $t \leq t'$ (M28), so $\xi(k)$ is non-decreasing in $t$ and non-increasing in $\mu$. With $\kappa$ fixed and $\mu' \leq \mu$, $k^*(\mu) \leq k^*(\mu')$ by M28, and

$$\xi\big(k^*(\mu);\mu\big) \leq \xi\big(k^*(\mu);\mu'\big) \leq \xi\big(k^*(\mu');\mu'\big).$$

With $s = -1$, $r$ uniform, $Q = 1$ and $k = 0$, $\xi(0) = \frac{1}{2}$ at $t = 0$ and 0 at $t = 1$, so $s \geq 0$ is needed.

□

## Proof of M32

*Lean:* `WeightSelection.winner_base_mono`, `WeightSelection.winner_base_mono_expect`, `WeightSelection.gain_half`, `WeightSelection.gain_two_thirds`, `WeightSelection.enters_at_half`, `WeightSelection.out_at_two_thirds`, `WeightSelection.selection_falls_with_exit`, `WeightSelection.winner_base_mono_needs_le_one`.

Write $a_k = s_k e_k$, $d_r = r_i - r_j$ and $d_a = a_i - a_j$. Maximality at $\mu_1$ and at $\mu_2$ gives

$$\mu_1 d_r + (1-\mu_1)d_a \geq 0, \qquad \mu_2 d_r + (1-\mu_2)d_a \leq 0.$$

Weighting the first by $1-\mu_2 \geq 0$ and the second by $1-\mu_1 \geq 0$ and subtracting,

$$(\mu_1 - \mu_2)\,d_r \geq 0, \qquad \text{so } r_i \leq r_j,$$

pointwise, and integrating gives the expectation. In the example, dividing by $\mu$ gives the entrant law $r+t$, $t = (1-\mu)/\mu$, and the outsider law $r$, so with $Q = 1$ and $m = 0$, by M1 and $\int F\,dF = \frac{1}{2}$,

$$\Delta(0) = \tfrac{1}{2} - \int_0^1 F_U(y-t)\,dy = \begin{cases} \frac{1}{2}, & t = 1\ (\mu = \frac{1}{2}), \\ \frac{1}{2} - \int_{1/2}^1 \big(y - \frac{1}{2}\big)\,dy = \frac{3}{8}, & t = \frac{1}{2}\ (\mu = \frac{2}{3}). \end{cases}$$

With the challenger in, both scores carry the bonus and the winner's base component is $\max(r_0, r_1)$; with her out, it is at most that maximum, and strictly below it on $\{r_0 < r_1 < r_0 + \frac{1}{2}\}$, where the incumbent wins with the lower base component, a set of positive measure. Above $\mu = 1$ the first order reverses.

□

**Proof of M33**

*Lean:* `Implications.varying_fee_reorders`, `Implications.cap_sets_count`, `Implications.cap_sets_count_model`, `Implications.compression_rises`, `Implications.compression_falls`, `Implications.compression_rises_needs_below`, `Implications.share_off_benchmark`, `Implications.share_ge_benchmark`.

With $u = \log$,

$$\kappa\big|_{w=3,\,c=2} = \log 3 > \log \tfrac{4}{3} = \kappa\big|_{w=2,\,c=1/2}.$$

If index $k$ entered with $(k+1)\kappa_k > V$, M10 would give $(k+1)\kappa_k \le V$, so index $k$ fails the entry condition and

$$k'^* \le k.$$

A compression $T$ is monotone and moves each wealth toward $x_0$ without crossing it, so $T \circ w$ is sorted and

$$w_{k^*-1} \le x_0 \Rightarrow w_{k^*-1} \le T(w_{k^*-1}), \qquad w_{k^*} \ge x_0 \Rightarrow T(w_{k^*}) \le w_{k^*},$$

and M12 gives the two signs. By M31, $\xi$ is non-increasing in $\mu$, and at $\mu = 1$ both laws are the law of $r$, so

$$\xi(k;\mu) \ge \xi(k;1) = \frac{k+1}{Q+1}.$$

$F = G$ gives $\xi(k) = \frac{k+1}{Q+1}$ (M31); the last statement is its contrapositive. $x \mapsto \frac{x+1}{2}$ compresses toward 1 and lowers 3, so the rise needs $w_{k^*-1} \le x_0$.

□

# B Evidence and verification

Every row of the model table is a theorem of Lean 4 with Mathlib. The Lean column names the theorems, and the proof of each row in Appendix A names the same ones. The development lives in `lean/mathlib/`, built on the toolchain `v4.24.0-rc1` against Mathlib commit `14871d5`, beside the core file `lean/EntryContest.lean`, which uses no Mathlib. The script `verify.sh` runs four kinds of check on every build, and `VERIFICATION.md` records what they printed.

1. *The axiom audit.* No source file contains `sorry`. Every declared theorem is passed to `#print axioms`. A Mathlib theorem that depends on an axiom other than `propext`, `Classical.choice` and `Quot.sound` fails the build, and a theorem of the core file that depends on an axiom other than `propext` and `Quot.sound` fails it too.

2. *The manuscript checker* (`checks/verify_manuscript.py`). Every model row names audited theorems and has exactly one proof naming the same theorems. Introduction rows point to model rows, and conclusion rows point to model or literature rows. Every literature quote appears verbatim in the cited paper's record under `lit/`. Plain English is at most 30% of every proof. Each rule also runs on a faulty sample, which the rule must reject.

3. *The literature records.* Each cited paper has a directory under `lit/` holding a reconstruction from the source, the claims our documents make about the paper with page locators, notes on every discrepancy found, and Lean for the mathematical claims. A record checks our reading of a source and does not reprove the source. The records run under `lit/verify_lit.sh`.

4. *The ledger of the conclusions* (`checks/ledger_literature.py`). The ledger declares the entities first, each a row of the measurement map with its table and its Observed code, the players among them with the options Table 2 gives them. It then records the conclusions as relations among the entities, a change in some entities that moves, pins, orders or caps another under conditions that are themselves entities, each citing the conclusions it represents, the rows they rest on and their standing against the papers read. The summary that opens Section 4 and Figure 1 are rendered from the ledger, so the prose carries no row identifiers. The ledger fails when an entity is not a row of the map, when a player's option is not in its row of Table 2, when a conclusion is represented by no relation, when a relation cites a row its conclusions do not rest on or a standing they do not state, when a relation with literature rows is not expressed exactly once, when a sentence does not name the entities of its relations, when a cited row has changed since the relations were last reviewed against it, and when the paragraph or the figure is not its rendering.

**What Lean does not carry.** The standing assumptions of Appendix D enter the theorems as hypotheses. The rows assume that $F$ and $G$ have no atoms rather than deriving it. The assumption follows from the primitives when $r$ has no atoms and $\mu \neq 0$ (`EntryContestRefute.investorLaw_noAtoms'`, `EntryContestRefute.nonInvestorLaw_noAtoms'`). A statement about what a cited paper says rests on a quotation with its page, recorded under `lit/`, and not on Lean.

**Illustrations.** The SymPy and numerical scripts under `checks/` still run in `verify.sh`, as illustrations and as regression tests. No row rests on them. Earlier drafts graded each claim by an evidence tier (Lean, SymPy or numerical) because several analytic steps entered the Lean development only as hypotheses. Those steps are now proved in Lean, and the tiers are retired.

# C Terminology

Every English term that carries load in a row or a proof is bound here to a symbol built from the primitives of Appendix D. Each entry names an object already defined, so the table is a translation key and not a second set of definitions. Indices follow Appendix A, with the richest challenger at index 0.

| Term | Symbol | Meaning |
|---|---|---|
| *Players and actions* | | |
| challenger | $i \in \{0,\dots,Q-1\}$ | A player of given wealth who chooses whether to compete. |
| incumbent | $C$ | The additional player, who competes in every case and appears only through her score CDF. Scored as an entrant, $C = F$: the open contest, in which the holder competes again under the rule that admits the challengers (Appendix D). Rows that hold for any atomless $C$ say so. Her entry is a parameter in M1 to M29 and a choice in M30. |
| scored as an entrant | $C = F$ | The incumbent's score has an entrant's law, as M9, M28, M30 and M31 take. |
| enter | $e_i = 1$ | Pay the fee $c$ and add $(1-\mu)s_i$ to the score. The only choice in the model; there is no effort. |
| entrant, outsider | $e_i = 1$, $e_i = 0$ | A challenger who enters, and one who does not. |
| entrant set | $S = \{i : e_i = 1\}$ | The challengers who enter. |
| prize | $V > 0$ | Fixed and awarded by rank, outside $u$. |
| *Wealth and cost* | | |
| wealth | $w_i$ | Challenger $i$'s resource. A challenger with $w_i \leq c$ does not enter. |
| wealth profile | $(w_0,\dots,w_{Q-1})$ | The realised wealths, common knowledge, sorted with $w_0 \geq w_1 \geq \cdots$. |
| entry cost | $\kappa(w) = u(w) - u(w-c)$ | The burden of the fee in utility units. |
| fee | $c$ | The nominal cost of entry, the same for every challenger. |
| burden-monotonicity | $\kappa$ strictly decreasing on $(c,\infty)$ | The assumption on $u$ (M23), weaker than strict concavity (M24). |
| *Scores and gains* | | |
| score | $\sigma_i = \mu r_i + (1-\mu)s_i e_i$ | The prize goes to the highest score. |
| winner's base component | $r_W$ | The base component of the player who takes the prize (M32). |
| common bonus | $s$ constant | A bought component equal for every holder, as Chiba City's four points are; it washes out when every player holds it (M32). |
| base component | $r_i$ | The part of the score every challenger has, whether or not she enters; drawn independently of wealth, of $s_i$ and across players, from an arbitrary law. |
| bought component | $s_i \geq 0$ | The part of the score that entry adds; drawn as $r_i$ is. Laws of $s_i$ that differ by identity are outside every row except M27 and M29. |
| weight on the base component | $\mu \in (0,1]$ | The rule's weight; $1-\mu$ weighs the bought component. "The rule weighs the base component more" means a larger $\mu$ (M28, M31). |
| score CDFs | $F$, $G$ | The laws of an entrant's score and an outsider's, with $F \leq G$ (M2). |
| dominance gap | $\varphi = G - F \geq 0$ | What entry buys. |
| best rival score | $H_m = CF^m G^{Q-1-m}$ | The CDF of the best score facing a challenger when $m$ others enter. |
| gain | $\Delta(m)$ | The value of entering when $m$ others enter, with $\Delta(m+1) \leq \Delta(m)$ (M4). |
| anonymity | $\Delta$ depends on $m$ alone | The gain depends on how many others enter and not on who they are (M26, M27). |
| first entrant's gain | $\Delta_Q(0)$, $D(Q)$ | The gain with no other entrant among $Q$ challengers, and its step in $Q$ (M16 to M22). |

| Term | Symbol | Meaning |
|---|---|---|
| *Equilibrium* | | |
| entry condition | $\kappa_j \le \Delta(j)$ | Index $j$ enters when its cost is at most its gain. |
| entry count | $k^*$ | The number of indices before the first one whose entry condition fails (M5). |
| count invariance | $\lvert S\rvert = k^*$ | Every pure-strategy equilibrium has $k^*$ entrants (M6). |
| identity multiplicity | $S \neq S'$, $\lvert S\rvert = \lvert S'\rvert = k^*$ | Different entrant sets can be equilibria at the same count (Appendix D). |
| assortative set | $\{0,\dots,k^*-1\}$ | The $k^*$ richest challengers (M7, M8). |
| marginal entrant | index $k^*-1$ | The poorest entrant of the assortative set. |
| first outsider | index $k^*$ | The richest challenger outside the assortative set. |
| *Comparative statics* | | |
| change of profile | $w \mapsto w'$ | Any map between sorted wealth profiles. No result assumes it keeps the mean unless it says so. |
| mean-preserving spread | $\sum_j w'_j = \sum_j w_j$, $w'$ majorizing $w$ | The case of economic interest, and not a hypothesis of M12. |
| displacement | $d_j = w'_j - w_j$ | The rank-by-rank difference of the two sorted profiles. |
| pivot-spread | $T$ with $T(w) \ge w$ for $w \ge x_0$ and $T(w) \le w$ for $w \le x_0$ | A non-decreasing map that spreads wealth on both sides of the pivot $x_0$ (M13). |
| ordering in dispersion | $z - y$ non-decreasing | For quantile functions $y$ and $z$, the second law is more dispersed [Lewis and Thompson, 1981]. With a crossing point, the transport map between the two laws is a pivot-spread. |
| margin condition | $d_{k^*-1} \ge 0$, or $d_{k^*} \le 0$ | The hypothesis of the rise branch, read at the marginal entrant, and of the fall branch, read at the first outsider (M12). |
| band | $d_{k^*-1} < 0 < d_{k^*}$ | The margins on which the margin condition is silent (M14). |
| *Aggregates* | | |
| entrants' share | $\xi(k^*)$ | The probability that the prize goes to an entrant, the incumbent included (M31). |
| total entry cost | $\sum_{i\in S} \kappa(w_i)$ | In utility units. Its resource counterpart $\lvert S\rvert c = k^* c$ is the same in every equilibrium. |
| sum of payoffs | $\sum_i U_i$ | Unweighted, so an interpersonal comparison of the index $u$ (M8). |
| selection | a rule picking one equilibrium | Not a refinement. Nothing claims that players coordinate on it. |

One name per object: an entrant is one who pays the fee (not a payer or an investor), the two parts of the score are the base and the bought component (not talent or ability), and $\mu$ is the weight on the base component. Words quoted from other papers in the literature table keep those papers' usage. Three pairs are kept apart because earlier drafts conflated them. The entry cost in utility, $\kappa$, varies with wealth, while the fee in resources, $c$, does not and is what society gives up. Count invariance holds, while identity multiplicity is real, and every comparative static concerns the count for that reason. A change of profile is what the results apply to, while the margin condition is what they assume.

# D Primitives and scope

**The contest.** One prize of value $V > 0$ is fixed and exogenous, and it is awarded by rank rather than sold. $Q$ challengers and one incumbent compete. Each challenger chooses $e_i \in \{0,1\}$, and entry costs the fee $c > 0$. The draws $r_i$ and $s_i \ge 0$ are independent of each other, across players and of the wealth profile, from arbitrary laws. The score is $\sigma_i = \mu r_i + (1-\mu) s_i e_i$, the highest score takes the prize, and the game is static and played once. "Equilibrium" therefore means the one-shot equilibrium of the entry game, with the incumbent's identity fixed, and K1's "every equilibrium" ranges over those; the claims hold at the state, the wealth profile and the holder, at which the game is played. A steady state, in which the winner holds the position next period, the displaced holder competes again, and wealth responds to past success, is outside the model. Each period of such a contest is the game studied here if players are myopic or value holding the position afterwards alike, since a common continuation value only enlarges the prize (M1); `STEADY_STATE_ADDENDUM.tex` sets out the rest. By M1 the gain is proportional to $V$, so $V$ enters only as a scale. Cole et al. [1992] read status as "a ranking device that determines how well he or she fares in the nonmarket sector", which is the sense of a prize awarded by rank.

**The two institutions.** Neither institution of the Research question is the model exactly. In Chiba City the four points are a fixed addition to the score, which fits the bought component as a common bonus (K9), and families without such care still compete on their other points; but the points are earned by using paid care for a threshold of

hours, at a price the provider sets, so the outlay is not one fixed fee. In Italian tenders a certified firm gains extra points and an uncertified firm still bids and can win, which fits the outsider who competes on the base component; but the certificate needs compliance with parameters set by decree as well as a payment, so the outlay buys standing only in part. The readings are AP-D1 and AP-D2 in `lit/applications_2026/CLAIMS.md`.

**The incumbent.** The incumbent is scored as an entrant, under the rule that admits the challengers. Her law is $C = F$, so she holds the bought component and differs from an entrant in nothing, and M26 includes her. This is the open contest, in which the holder competes again under the same rule. Her entry is a parameter in M1 to M29. In M30 it is a choice, she is player 0 of a game of $Q+1$ players with the lowest cost, and outside the band of M6 she is in every non-empty equilibrium, so $C = F$ is the equilibrium outcome there. Two institutions are not modelled, and this choice turns them off. A protected holder, who keeps the position unless the best challenger clears a fixed bar, is a law $C$ that stays put as $\mu$ varies. A holder with an edge beyond wealth is a law above $F$, and sits at the boundary of M27. Every row holds for an arbitrary atomless $C$ except M9, M20, M22, M28 and M31, which take $C = F$, M29, whose witness fixes $C$, M30, where her entry is chosen, and M21, which needs $C(a) > 0$. A fixed $C$ therefore keeps the count, the sorting, the cap and the spread results, and loses the two $\mu$ benchmarks and the directions in $\mu$ of M28 and M31.

**Payoffs.** With $u$ the utility of wealth,

$$U_i = u(w_i - c\,e_i) + V\,\mathbf{1}\{i \text{ wins}\}.$$

The fee is paid inside $u$, so its burden is felt in utility and varies with wealth. The prize enters outside $u$ and linearly, so rich and poor value it alike. Inequality therefore acts through one channel, the cost of entry. This is the privilege contest of Schroyen and Treich [2016], and not their rent-seeking contest, in which a money prize would sit inside $u$.

**After entry.** The model stops at entry. One prize is awarded by rank, and nothing after the award is modelled. Differences in pay among those who hold positions, such as the class pay gap of Laurison and Friedman [2016], are outside it; the comparable quantity is the share of positions held by entrants (M31).

**No effort margin.** Entry costs a fixed fee and buys a fixed component of the score, and there is no effort to choose. A more accurate ranking of that component therefore does not lower what entry buys, and the count does not fall as the component weighs more (M28). With an effort choice the sign is the opposite (L20, L22). In Morgan et al. [2022] a more accurate ranking raises the marginal return to effort, effort rises until competing earns nothing, and the prizes then clear only if some contestants drop out. With logistic noise of scale $\sigma$, cost $k\,e^{\alpha x}$ in log-output and a mass $m$ of prizes, their equilibrium conditions give participation $m/(1-\sigma\alpha)$ in the drop-out region, because the logistic hazard equals the distribution function. Participation rises with the noise, from $m$ as $\sigma \to 0$ to 1 at $\sigma = (1-m)/\alpha$, and their second-order condition holds throughout (`MeritocracyLogistic.mixed_region`, `MeritocracyLogistic.participation_strictMonoOn`, `MeritocracyLogistic.soc_mixed`). This instance is derived here in their model, and they do not state it. In the free-entry extension of Drugov and Ryvkin [2020] the less dispersed noise brings higher effort and weakly fewer entrants, the same sign for the same reason (L22).

**The institution's weight.** The weight $\mu$ is given, and the model supplies one side of an institution's choice of it. With the fee, the prize and the wealth profile fixed, a rule that weighs the base component more draws no more entrants (M28) and gives them no larger share of the prize (M31). At $\mu = 1$ entry buys nothing: every score is the base component, the two score laws coincide, and the gain of M1 is zero. The other side, how the base component of the winner moves with $\mu$, is signed only with the entrants fixed, where it does not fall (M32). Once entry responds it can fall: a common bonus washes out when every challenger holds it, and favours its holders once some stop buying (M32). So the model does not say where an institution sets its weight. The motivation is the one of the Research question: a credential is a cleaner status, cheap to check, than the disorganised status that comes from unverified providers. A model in which the institution chooses $\mu$ against that information is left to further work; `WEIGHT_ADDENDUM.tex` sets out what it needs.

**Utility.** $u$ is strictly increasing and burden-monotone, so $\kappa(w) = u(w) - u(w-c)$ is strictly decreasing on $(c, \infty)$. Strict concavity implies it (M23), and it is strictly weaker (M24). One $u$ serves every challenger, so $\kappa$ varies across challengers through wealth alone. $u$ is not observed. It represents the recorded entries rather than determining them, preference being "the binary representation of individual choice" [Sen, 1973, p. 253] and utility being "used to represent preferences" [Schoemaker, 1982, p. 532], and it enters the model only through $\kappa$. $\kappa$ is a difference of such a $u$ and not a measure of how strongly the fee is felt [Schoemaker, 1982, p. 533]. The statement that $\kappa$ falls with wealth therefore has no content of its own. Its content is what it implies for a record of wealths, fee, prize and entries. The richest $k^*$ challengers entering is an equilibrium, and the only one when $\Delta(k^*-1) < \kappa_{k^*}$ (M7, M8), and the count moves with the prize, the fee and the profile as M11 to M13 state. A challenger whose wealth is at or below $c$ does not enter. The strictness result M15 also uses $\kappa(w) \to \infty$ as $w \downarrow c$, which holds exactly when $u(x) \to -\infty$ as $x \downarrow 0$ (M25).

**Score laws.** $F$ is the law of $\mu r + (1-\mu)s$ and $G$ the law of $\mu r$, and both are taken to have no atoms (Appendix B). Earlier drafts also assumed a common lower endpoint for $F$ and $G$ and that $\varphi$ vanishes at both ends of the support. No row uses either, and both are dropped.

**Information.** The realised wealth profile is common knowledge, entry is simultaneous, and the draws $r_i$ and $s_i$ are made after entry. At the entry stage a challenger knows $c$, $V$, $\mu$, the laws of $r$ and $s$ and the wealth profile, and not her own draws, which reach her decision only through $F$ and $G$. Her condition $\kappa(w_i) \le \Delta(j)$ has $w_i$ as its only individual datum, so neither score component enters the cost of entry or the decision. Because the profile is common knowledge, the entry condition is written on the sorted profile, and $k^*$ is a count shared by every pure-strategy equilibrium (M6). It does not pin who enters. With $\kappa = (2,5,8)$ and $\Delta = (12,10,1)$, the sets $\{0,1\}$, $\{0,2\}$ and $\{1,2\}$ are all equilibria, the last without the richest challenger (`EntryContestEq.three_equilibria`). Identities are pinned when $\Delta(k^*-1) < \kappa_{k^*}$ (M7). Under private entry costs the count would be binomial and the entrant set random, as in Moreno and Wooders [2011], where every buyer compares her own cost with one threshold. Here the threshold depends on the index, $\kappa_j \le \Delta(j)$, and it is flat only when $\Delta$ is constant, which M4 rules out whenever $\int \varphi^2 \, dK_m > 0$.

**The anonymity boundary.** Independence of the draws from wealth makes $F$ and $G$, and so $\Delta$, depend on the count of entrants alone (M26). Economies in which the law of the bought component varies with wealth are outside every row except M27 and M29, which mark the boundary. The exclusion is shown and not only asserted. When the bought component falls with wealth, equilibria of different sizes coexist (M27), and count invariance fails. It also fails for some costs when it rises with wealth (M29). The question of how many enter and the question of which enter then cease to be separable, and every comparative static of the model concerns the count.

# E Refuted conjectures

**R1, a collapse threshold.** An earlier draft conjectured a finite $\bar{Q}$ beyond which $k^* = 0$. The conjecture is false. With $r$ uniform on $[0,1]$, $s = 0$ with probability $p$ and $s = 1$ otherwise, $0 < \mu \le \frac{1}{2}$ and the incumbent scored as an entrant, for every $Q \ge 1$

$$\Delta_Q(0) = V\Big(\frac{1-p^2}{2} - \frac{p(1-p)}{Q+1}\Big) \ge V\,\frac{1-p}{2}, \qquad \Delta_Q(0) \to V\,\frac{1-p^2}{2},$$

so a challenger whose cost is at most $V(1-p)/2$ enters for every $Q$ (`EntryContestRefute.bern_Delta`, `EntryContestRefute.bern_Delta_lower`, `EntryContestRefute.bern_Delta_limit`). When an outsider's score never exceeds $x_G$, $\Delta(m) \ge V\big(\int_{x \ge x_G} CF^m \, dF - \frac{1}{Q-m}\big)$ for $m < Q$, and the first term does not depend on $Q$ (`EntryContestRefute.r1_lower_bound`). An earlier draft gave the limit as $\Pr(X > \mu)$ times the chance of beating the incumbent. That formula is withdrawn, since in the family above $\Pr(X > \mu) = 1 - p$ (`EntryContestRefute.bern_investor_above`) while the limit is $V(1-p)(1+p)/2$.

**R2, fewer entrants as $Q$ grows.** A referee conjectured that for $\mu > 0$ a larger $Q$ shrinks the gain, so fewer enter. The conjecture is not general. In the family of R1 with $0 < p < 1$, $\Delta_Q(0)$ rises strictly at every $Q$ (`EntryContestRefute.bern_Delta_strictMono`), so a challenger whose cost lies between $\Delta_Q(0)$ and $\Delta_{Q+1}(0)$

stays out with $Q$ challengers and enters with $Q+1$. With $r$ uniform the gain never falls in $Q$ (M18). The conjectured direction occurs for other primitives (M20), and the sign of the effect is set by the law of $r$ (M16 to M22).

# F Responses to the referee reports

**Asymmetric $k$-of-$Q$ equilibria.** Both referees found that the earlier version compared aggregate participation regimes instead of checking unilateral deviations. The model now derives the asymmetric equilibrium in which $k^*$ of the $Q$ challengers enter, checks deviations index by index, and shows the count is the same in every equilibrium (M5 to M8).

**Prize and affordability.** The second referee objected that the earlier version used one quantity both as inequality and as the prize, which made the inequality result close to tautological, and that a larger prize should raise every participant's incentive. The prize is now fixed and separate from the wealth distribution (Appendix D). A larger prize raises every gain and does not lower the entry count, and a larger fee does not raise it (M11). The inequality result is the sign rule of M12 to M15, in which $V$ does not move.

**The mobility ceiling.** Both referees found the old ceiling an artefact of uniform supports and of a limit in $Q$ under which the equilibrium does not exist. The saturation bound replaces it. An entrant at index $m$ with cost $\kappa$ has $(m+1)\kappa \le V$, so the marginal entrant gives $k^* \le V/\kappa_{k^*-1}$ for every $Q$, with no limit taken (M10).

# G Coverage of the referee comments

The two reports on the earlier submission (BEJEAP, rejected 31 August 2026) are recorded comment by comment in `HANDOVER_referee_reports_20260831.md`. Each comment is marked *answered* when a model row or an appendix settles it, *partly answered* when part of it remains, *open* when nothing here settles it, and *moot* when it concerned text or an object that the rebuilt model no longer has. Comments that only praised a change are omitted.

| Comment | Substance | Status | Where |
|---|---|---|---|
| *Reviewer 1* | | | |
| R1.1 | Aggregate regimes were compared in place of unilateral deviations. | answered | Deviations are checked index by index. The count $k^*$ is the same in every pure-strategy equilibrium, and the assortative set is one (M5 to M8). |
| R1.2 | The reach into education policy is limited by a zero-sum, fixed-slot contest. | answered | The rebuilt model makes no policy claim. The prize is fixed and awarded by rank, and the scope is stated with it (Appendix D). |
| R1.3 | Results were asserted beyond uniform distributions without proof. | answered | Every row holds for general laws of $r$ and $s$ unless it names a family or a class, and the rows that do say so (M18 to M22, M27). |
| R1.4 | The contribution is not located against contests with entry costs, positional education and status competition. | answered | The literature table (L1 to L33) places each result. L14 to L16 place the status reading against Weber's distinction of class from status, and L17 to L19 against access to positions, closure and opportunity hoarding (C10). The novelty reading is complete, and every claim that something is new rests on papers read in full (Table 5). |
| R1.4 | The mobility ceiling reflects the uniform specification. | answered | The ceiling is replaced by a bound for every $Q$ and every law, with no limit taken (M10). |
| R1.a2 | "Stable Nash equilibrium" is undefined. | answered | Only pure-strategy Nash equilibrium is used (M6). |
| R1.a3 | Full withdrawal should be a third regime. | answered | The count $k^*$ ranges over $0,\dots,Q$, so no entry, partial entry and full entry are all cases of one result (M5, M6). |
| R1.a4 | The ceiling is used inconsistently across regimes. | answered | One bound covers every count (M10). |
| R1.a5 | Prize effects and affordability effects are conflated. | answered | The prize sits outside $u$ and the fee inside it (Appendix D). The prize and the fee move the count separately (M11). |
| R1.a6 | The notation $\Phi(Q)$ is inconsistent. | moot | The rebuilt model has no $\Phi$. |
| R1.a7 | Comparative statics in $\mu$ were misstated. | open | No comparative static in $\mu$ is claimed. |
| R1.a8 | The paper is verbose. | answered | Results sit in tables, and every proof is at most 30% prose, checked by script (Appendix B). |

| Comment | Substance | Status | Where |
|---|---|---|---|
| *Reviewer 2* | | | |
| R2.1 | Asymmetric equilibria in which only some challengers enter. | answered | The equilibrium is the count $k^*$ of entrants, and every equilibrium has it (M5, M6). |
| R2.1 | More enter as the prize rises and as the fee falls. | answered | M11. |
| R2.1 | More enter as social capital matters more ($\mu$ falls). | answered | With the incumbent scored as an entrant, the gain and the count are non-increasing in $\mu$ for every law of $r$ and of $s \geq 0$ (M28). |
| R2.1 | At $\mu = 0$ the number entering does not depend on $Q$. | answered | At $\mu = 0$ the gain is $V/(m+2)$ for every $Q$ (M9), so with the same costs the count is the same. |
| R2.1 | At $\mu > 0$ a larger $Q$ means fewer enter. | answered | Not in general. The first entrant's gain can rise in $Q$, never falls with a uniform base component, and the sign depends on the law of $r$ (M16 to M22, conjecture R2 in Appendix E). |
| R2.1 | An equilibrium in which the incumbent abstains and a challenger enters. | answered | Only inside the band. With the incumbent as player 0 of a game of $Q+1$ players, such an equilibrium exists iff $g(k^*) < \kappa_0$ and $\kappa_{k^*} \leq g(k^*-1)$, and the assortative set, with her in, is an equilibrium in every case (M30). |
| R2.2 | The effect of inequality on status expenditure is unclear, and the prize is the inequality. | answered | The prize is fixed and apart from the wealth distribution. A spread changes the count, and so the resource expenditure $k^*c$, with a sign set by the marginal entrant against the pivot (M12 to M15). |
| R2.2 | Endowment inequality should reduce status expenditure, as in the literature. | answered | It does when the marginal entrant is at or below the pivot, and the reverse holds when it is above (M13). |
| R2.3 | Policy is discussed without an objective. | partly answered | No policy claim is made. Among equilibria of one size the assortative set has the least total utility cost, and the resource cost $k^*c$ is the same in all of them (M6, M8). No welfare section is attempted. |
| R2.4 | The ceiling is an artefact, inconsistent with equilibrium, and set in an artificial comparative static. | answered | M10 bounds $k^*$ for every $Q$ with no limit taken. |
| R2.5 | A static one-shot model would be simpler. | answered | The game is static and played once (Appendix D). |
| R2.5 | Give the incumbent a lower cost instead of relying on concavity. | answered | M5 to M8 hold for any ordered costs, whatever their source, and concavity is replaced by burden-monotonicity (M23, M24). |
| R2.5 | Financial constraints are not crucial. | answered | No budget constraint is used beyond a challenger at or below the fee not entering (Appendix D). |
| R2.5 | Consider a reduced-form contest success function. | partly answered | The score is kept, and every result is stated in the laws $F$, $G$ and $C$ of the scores. |
| R2.5 | Why is $\mu < \frac{1}{2}$ needed? | answered | No row assumes it. The family of conjecture R1 uses $\mu \leq \frac{1}{2}$ only to be explicit. |
| R2.6 | Results should be precise propositions with proofs. | answered | Each model row states hypotheses and a conclusion and has a proof checked in Lean (Appendix A). |
| R2.6 | Errors in the text of the earlier manuscript (saving, $p_1$, marginal utility cost, income ratios, free riding, the Sutton Trust passage). | moot | The text is retired. One point carries over. A uniform rise in every wealth lowers every cost, so the count does not fall (M12). |

# References


Stephen Coate and Glenn C. Loury. Will affirmative-action policies eliminate negative stereotypes? *American Economic Review*, 83(5):1220–1240, 1993.

Harold L. Cole, George J. Mailath, and Andrew Postlewaite. Social norms, savings behavior, and growth. *Journal of Political Economy*, 100(6):1092–1125, 1992.

William J. Corcoran. Long-run equilibrium and total expenditures in rent-seeking. *Public Choice*, 43(1):89–94, 1984. doi: 10.1007/BF00137909.

William J. Corcoran and Gordon V. Karels. Rent-seeking behavior in the long-run. *Public Choice*, 46(3):227–246, 1985. doi: 10.1007/BF00124421.

Robert M. Costrell and Glenn C. Loury. Distribution of ability and earnings in a hierarchical job assignment model. *Journal of Political Economy*, 112(6):1322–1363, 2004.

Mikhail Drugov and Dmitry Ryvkin. How noise affects effort in tournaments. *Journal of Economic Theory*, 188: 105065, 2020.

Raquel Fernández and Jordi Galí. To each according to...? markets, tournaments, and the matching problem with borrowing constraints. *Review of Economic Studies*, 66(4):799–824, 1999.

Roland G. Fryer and Glenn C. Loury. Categorical redistribution in winner-take-all markets. Working Paper 10104, National Bureau of Economic Research, 2003.

Roland G. Fryer and Glenn C. Loury. Valuing identity. Working Paper 16568, National Bureau of Economic Research, 2010.

Roland G. Fryer and Glenn C. Loury. Valuing diversity. *Journal of Political Economy*, 121(4):747–774, 2013. doi: 10.1086/671180.

Qiang Fu and Jingfeng Lu. Contest with pre-contest investment. *Economics Letters*, 103(3):142–145, 2009. doi: 10.1016/j.econlet.2009.03.011.

Qiang Fu and Jingfeng Lu. Contest design and optimal endogenous entry. *Economic Inquiry*, 48(1):80–88, 2010. doi: 10.1111/j.1465-7295.2008.00135.x.

Qiang Fu, Qian Jiao, and Jingfeng Lu. Contests with endogenous entry. *International Journal of Game Theory*, 44 (2):387–424, 2015. doi: 10.1007/s00182-014-0435-9.

R. L. Fullerton and R. P. McAfee. Auctioning entry into tournaments. *Journal of Political Economy*, 107:573–605, 1999.

Refael Hassin and Moshe Haviv. Equilibrium threshold strategies: The case of queues with priorities. *Operations Research*, 45(6):966–973, 1997. doi: 10.1287/opre.45.6.966.

Ed Hopkins and Tatiana Kornienko. Which inequality? the inequality of endowments versus the inequality of rewards. *American Economic Journal: Microeconomics*, 2(3):106–137, 2010. doi: 10.1257/mic.2.3.106.

Lily Hu, Nicole Immorlica, and Jennifer Wortman Vaughan. The disparate effects of strategic manipulation. In *Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT* '19)*, pages 259–268. ACM, 2019. doi: 10.1145/3287560.3287597.

Daniel Laurison and Sam Friedman. The class pay gap in Britain's higher professional and managerial occupations. *American Sociological Review*, 81(4):668–695, 2016. doi: 10.1177/0003122416653602.

Edward P. Lazear and Sherwin Rosen. Rank-order tournaments as optimum labor contracts. *Journal of Political Economy*, 89(5):841–864, 1981.

Dan Levin and James L. Smith. Equilibrium in auctions with entry. *American Economic Review*, 84(3):585–599, 1994.

Toby Lewis and J. W. Thompson. Dispersive distributions, and the connection between dispersivity and strong unimodality. *Journal of Applied Probability*, 18(1):76–90, 1981.

Lydia T. Liu, Nikhil Garg, and Christian Borgs. Strategic ranking. In *Proceedings of the 25th International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 151 of *Proceedings of Machine Learning Research*, pages 2489–2518, 2022. arXiv:2109.08240.

Timothy Mathews and Soiliou Namoro. Participation incentives in rank order tournaments with endogenous entry. *Journal of Economics*, 95(1):1–23, 2008. doi: 10.1007/s00712-008-0017-z.

Diego Moreno and John Wooders. Auctions with heterogeneous entry costs. *RAND Journal of Economics*, 42(2): 313–336, 2011.

John Morgan, Henrik Orzen, and Martin Sefton. Endogenous entry in contests. *Economic Theory*, 51(2):435–463, 2012.

John Morgan, Justin Tumlinson, and Felix Várdy. The limits of meritocracy. *Journal of Economic Theory*, 201: 105414, 2022. doi: 10.1016/j.jet.2022.105414.

Dmitry Ryvkin and Mikhail Drugov. The shape of luck and competition in winner-take-all tournaments. *Theoretical Economics*, 15(4):1587–1626, 2020. doi: 10.3982/TE3824.

Paul J. H. Schoemaker. The expected utility model: Its variants, purposes, evidence and limitations. *Journal of Economic Literature*, 20(2):529–563, 1982.

Fred Schroyen and Nicolas Treich. The power of money: Wealth effects in contests. *Games and Economic Behavior*, 100:46–68, 2016.

Amartya Sen. Behaviour and the concept of preference. *Economica, New Series*, 40(159):241–259, 1973.

Moshe Shaked. Dispersive ordering of distributions. *Journal of Applied Probability*, 19(2):310–320, 1982.

Charles Tilly. *Durable Inequality*. University of California Press, Berkeley, 1998.

Max Weber. *Economy and Society: An Outline of Interpretive Sociology*. University of California Press, Berkeley, 1978.

Bernd Wegener. Concepts and measurement of prestige. *Annual Review of Sociology*, 18:253–280, 1992. doi: 10.1146/annurev.so.18.080192.001345.